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Zeituni et al.

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(54) **IMAGE SENSOR ARRAY WITH CAPACITIVE CURRENT SOURCE AND SOLID-STATE IMAGING DEVICE COMPRISING THE SAME**

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CPC ***H04N 25/771*** (2023.01); ***H04N 25/7795*** (2023.01); ***H04N 25/78*** (2023.01); *H04N 25/616* (2023.01)

(58) **Field of Classification Search**
CPC H04N 25/616; H04N 25/771; H04N 25/7795; H04N 25/78
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2010/0207920 A1 8/2010 Chaji et al.
2012/0194715 A1 8/2012 Skaug
(Continued)

OTHER PUBLICATIONS

Brom et al., "Ramped Capacitor Current Source With Software Controlled Non-Linearity Compensation", Precision Electromagnetic Measurements Digest, 2004 Conference on, IEEE, Jun. 27-Jul. 2, 2004, pp. 66-67.
(Continued)

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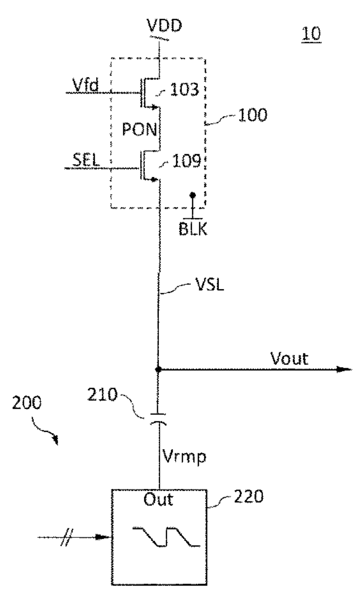
(57) **ABSTRACT**

An image sensor array includes a pixel circuit that generates a pixel output signal, wherein an amplitude of the pixel output signal is related to an intensity of detected light. The pixel circuit passes the pixel output signal to a data signal line for a selection period. A current control capacitor supplies a current to the data signal line through a first electrode in the selection period. A ramp generator generates a voltage ramp signal and passes the voltage ramp signal to a second electrode of the current control capacitor in the selection period.

15 Claims, 17 Drawing Sheets



(51) **Int. Cl.**

H04N 25/78	(2023.01)
H04N 25/616	(2023.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

2014/0209811	A1	7/2014	Chou	
2015/0208008	A1	7/2015	Gendai	
2017/0141685	A1*	5/2017	Cannankurichi	H02M 3/1563
2018/0013294	A1*	1/2018	Edelson	H03K 17/162
2018/0152017	A1*	5/2018	Langer	H02H 9/025
2018/0191983	A1	7/2018	Wane et al.	
2019/0123089	A1	4/2019	Doege	
2020/0003874	A1	1/2020	Moriyama	
2020/0260031	A1	8/2020	Ebihara et al.	

OTHER PUBLICATIONS

International Search Report and Written Opinion mailed on Jun. 29, 2022, received for PCT Application PCT/ EP2022/057484, filed on Mar. 22, 2022, 12 pages.

* cited by examiner

FIG 1

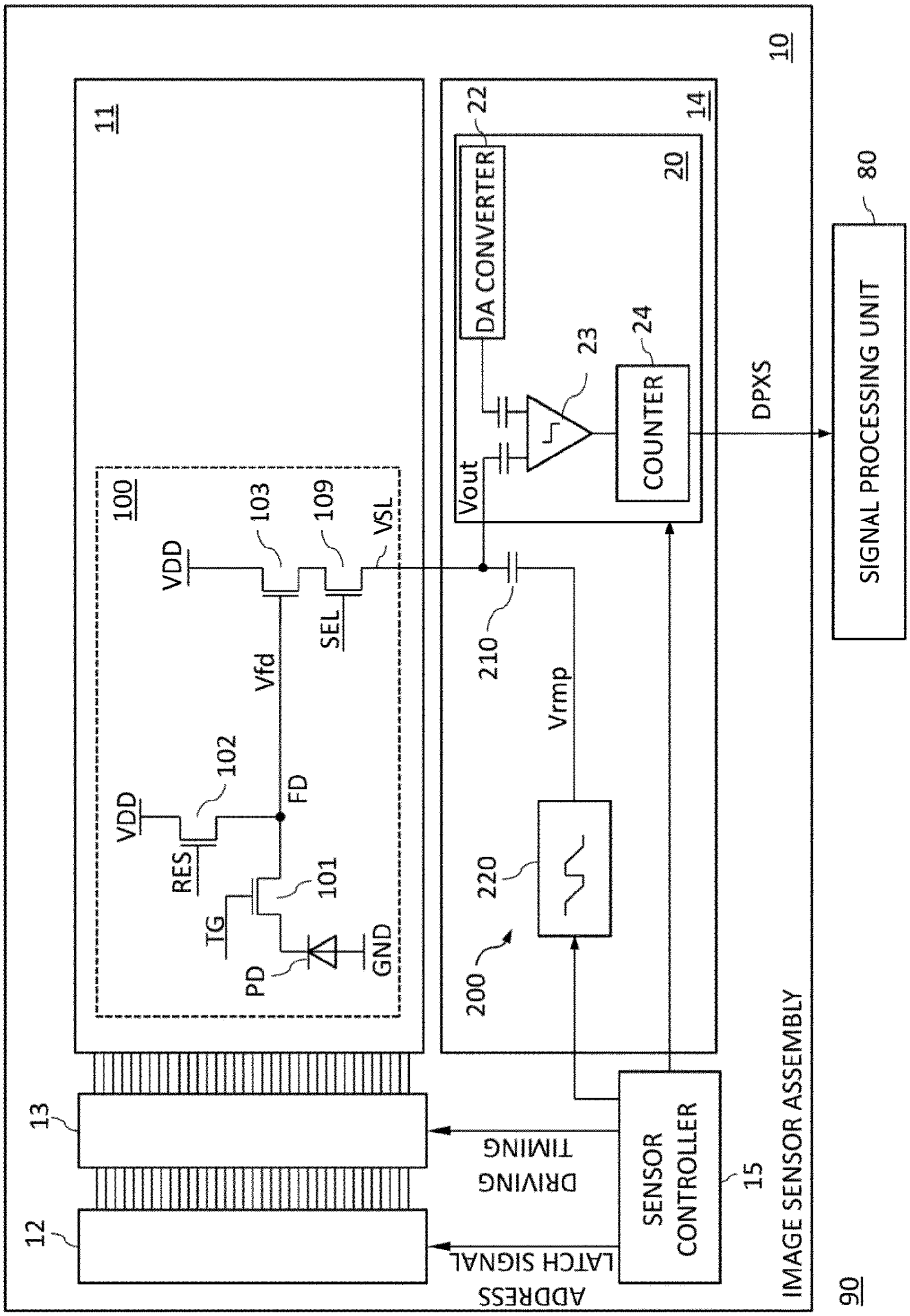
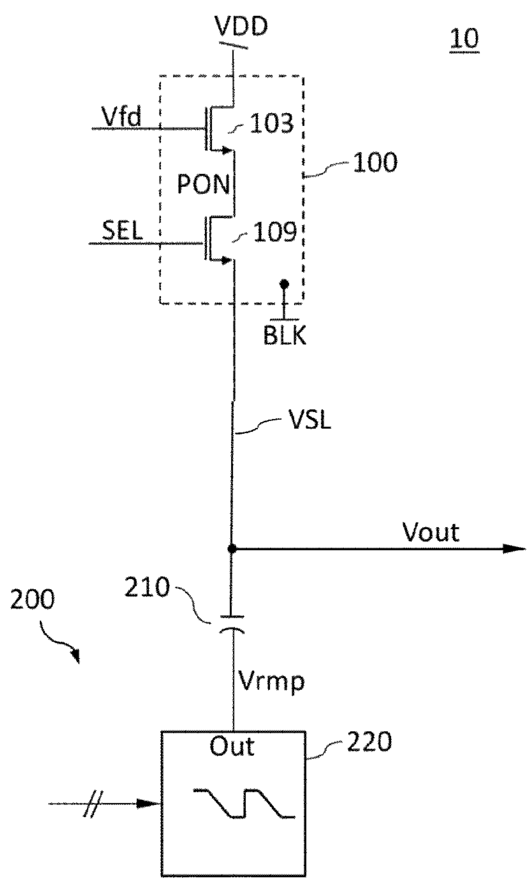
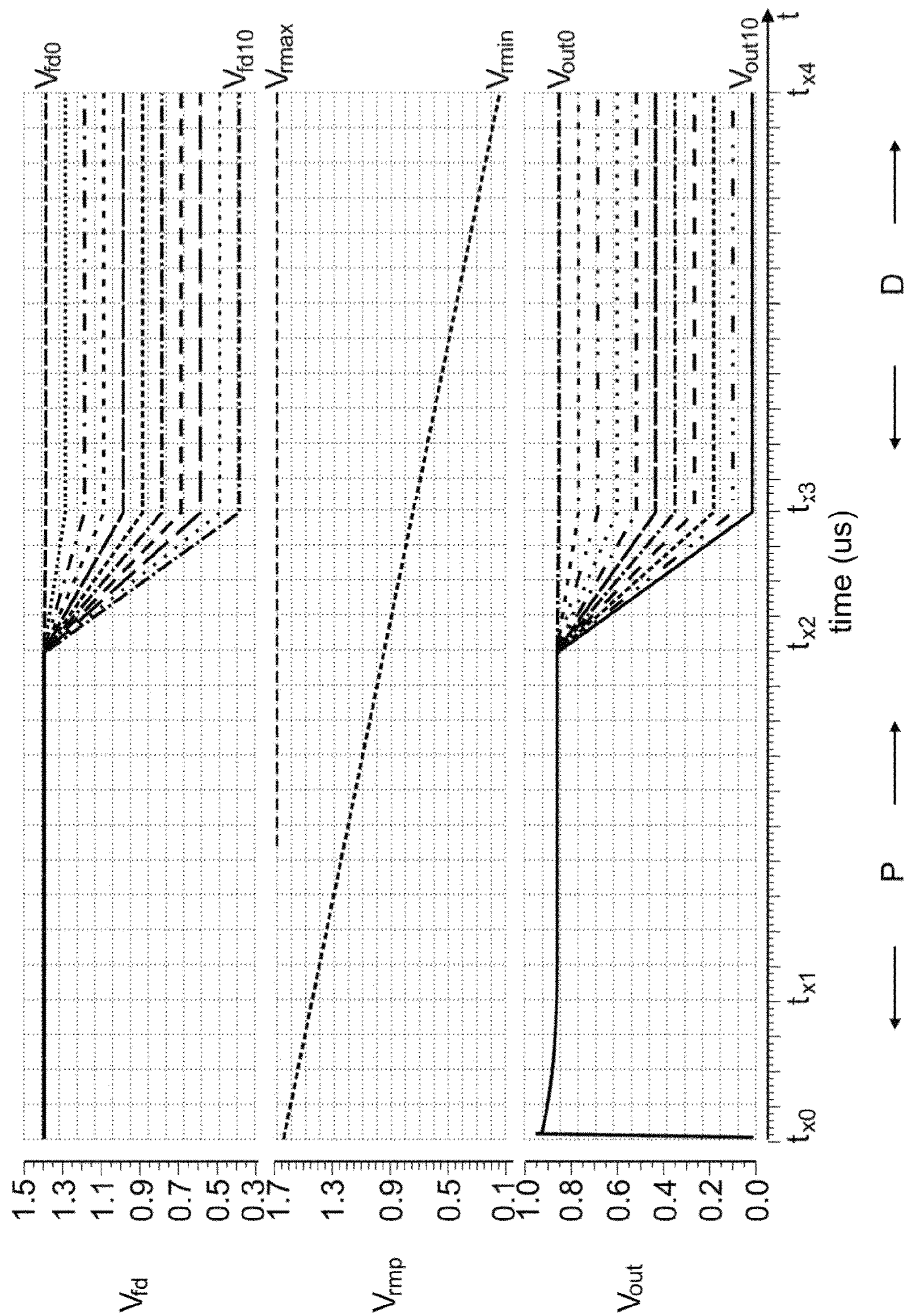

10
11
100
PD
GND
TG
101
RES
VDD
102
FD
Vfd
VDD
103
109
SEL
VSL
14
20
22
DA CONVERTER
23
24
COUNTER
Vout
210
Vrmp
200
220
DPXS
80
SIGNAL PROCESSING UNIT
12
13
ADDRESS
LATCH SIGNAL
DRIVING
TIMING
SENSOR
CONTROLLER
15
IMAGE SENSOR ASSEMBLY
90

FIG 2





V_{fd}
V_{rmp}
V_{out}
1.5
1.3
1.1
0.9
0.7
0.5
0.3
1.7
1.3
0.9
0.5
0.1
1.0
0.8
0.6
0.4
0.2
0.0
V_{fd0}
V_{fd10}
V_{rmax}
V_{rmin}
V_{out0}
V_{out10}
t
t_{x0}
t_{x1}
t_{x2}
t_{x3}
t_{x4}
time (us)
P
D

Fig. 3

FIG 4

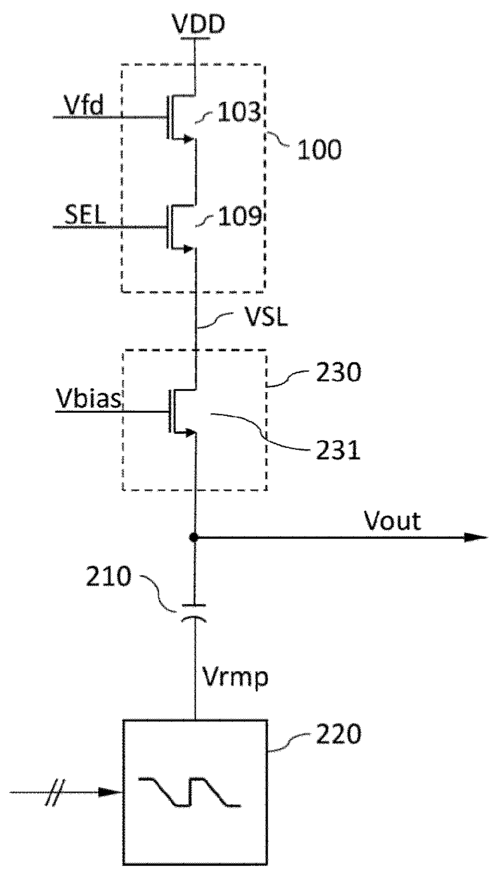
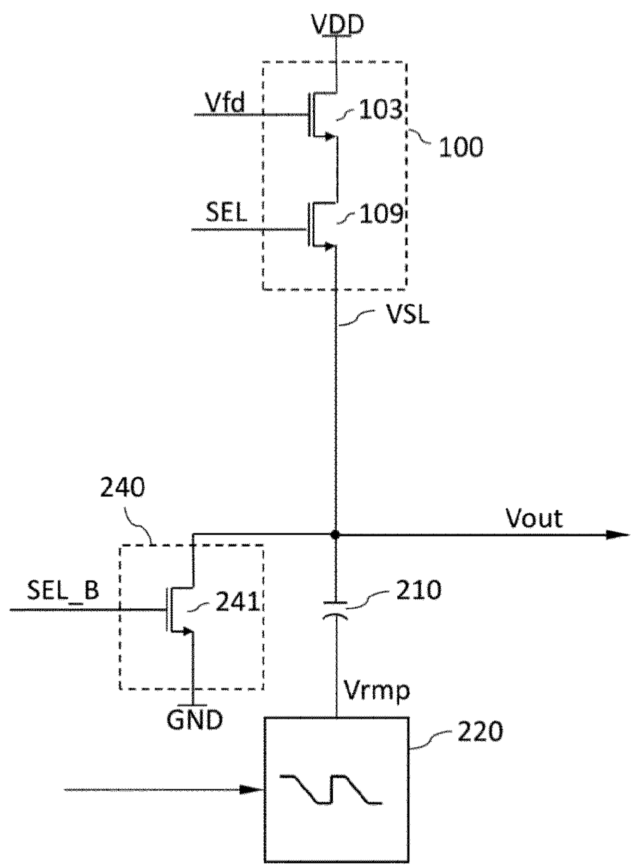


FIG 5



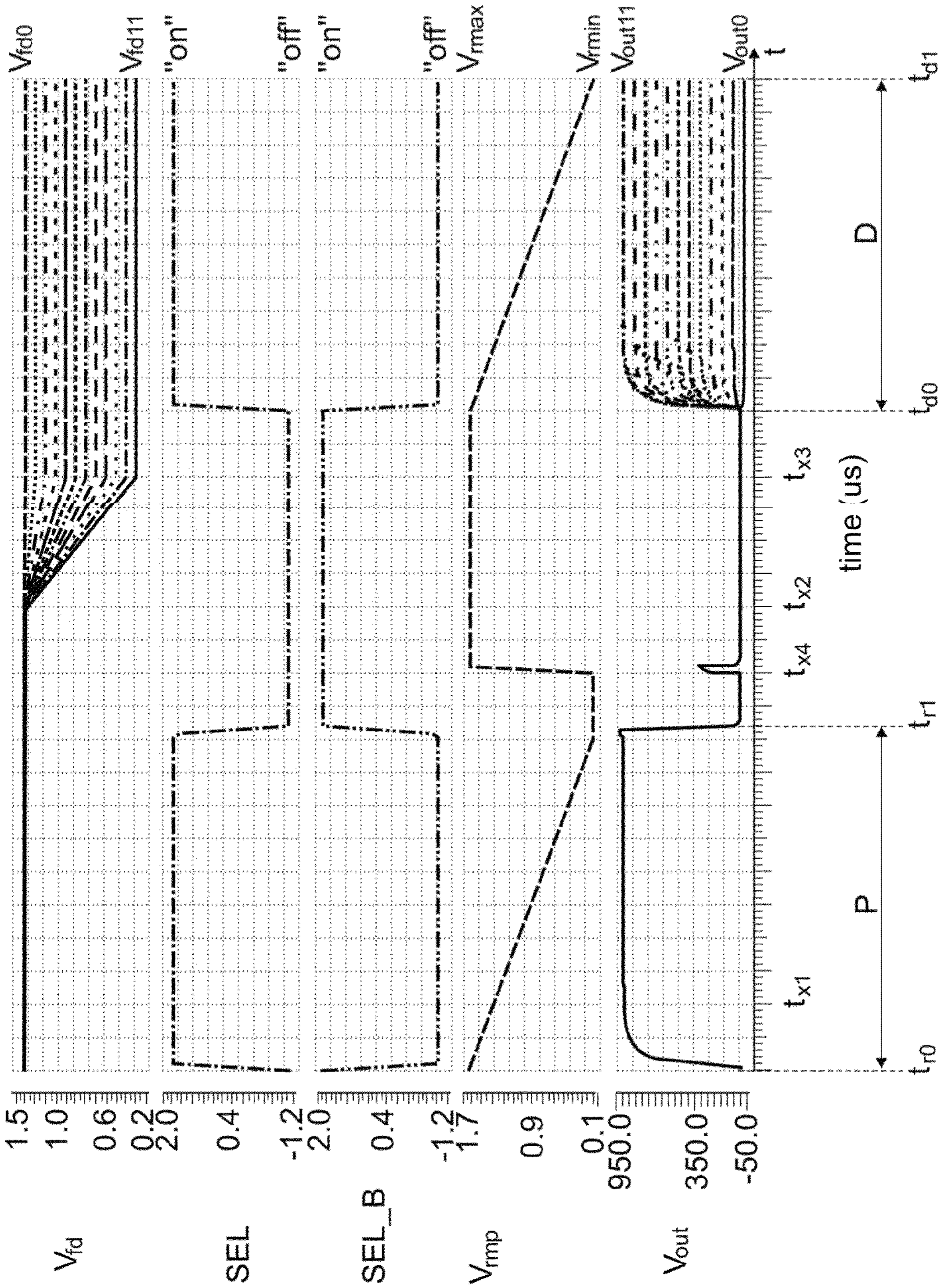

V_{fd}
1.5
1.0
0.6
0.2
V_{fd0}
V_{fd11}
SEL
2.0
0.4
-1.2
"on"
"off"
SEL_B
2.0
0.4
-1.2
"on"
"off"
V_{rmp}
1.7
0.9
0.1
V_{rmax}
V_{rmin}
V_{out}
950.0
350.0
-50.0
V_{out11}
V_{out0}
t
t_{r0}
t_{x1}
P
t_{r1}
t_{x4}
t_{x2}
t_{x3}
t_{d0}
D
t_{d1}
time (us)

Fig. 6

FIG 7

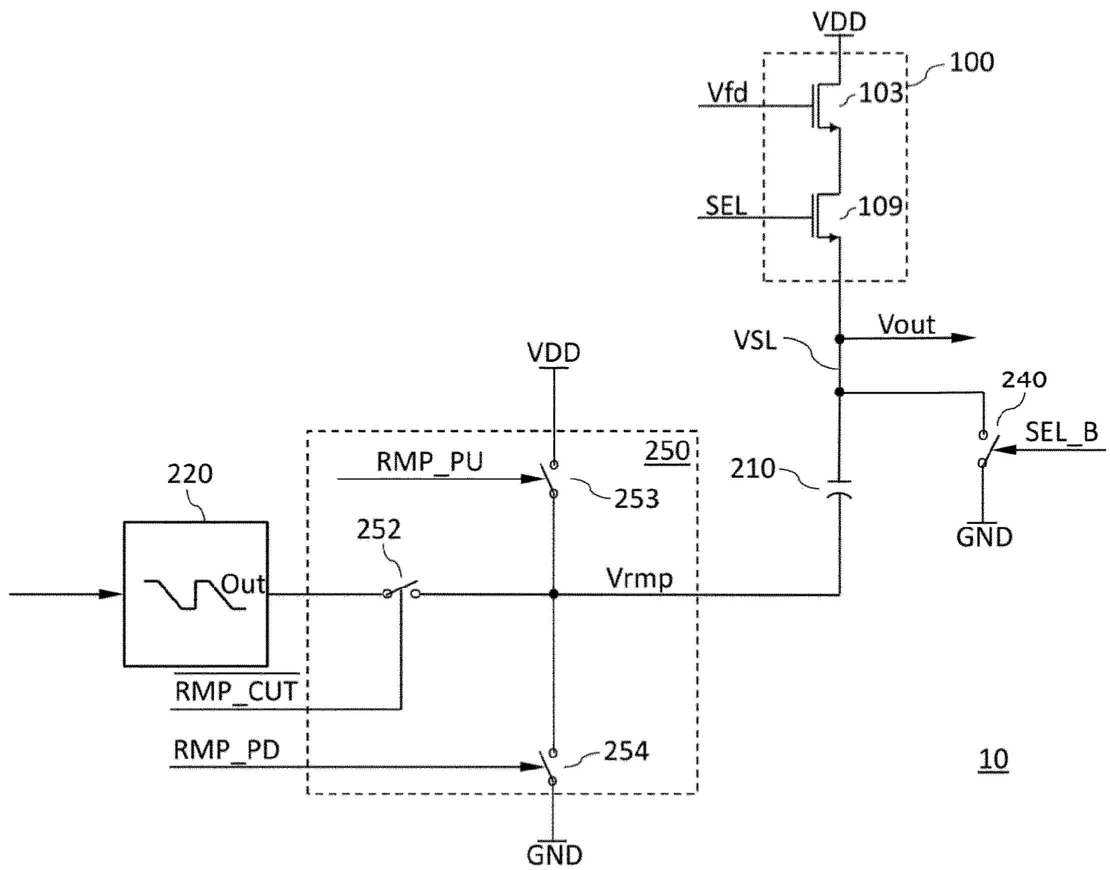
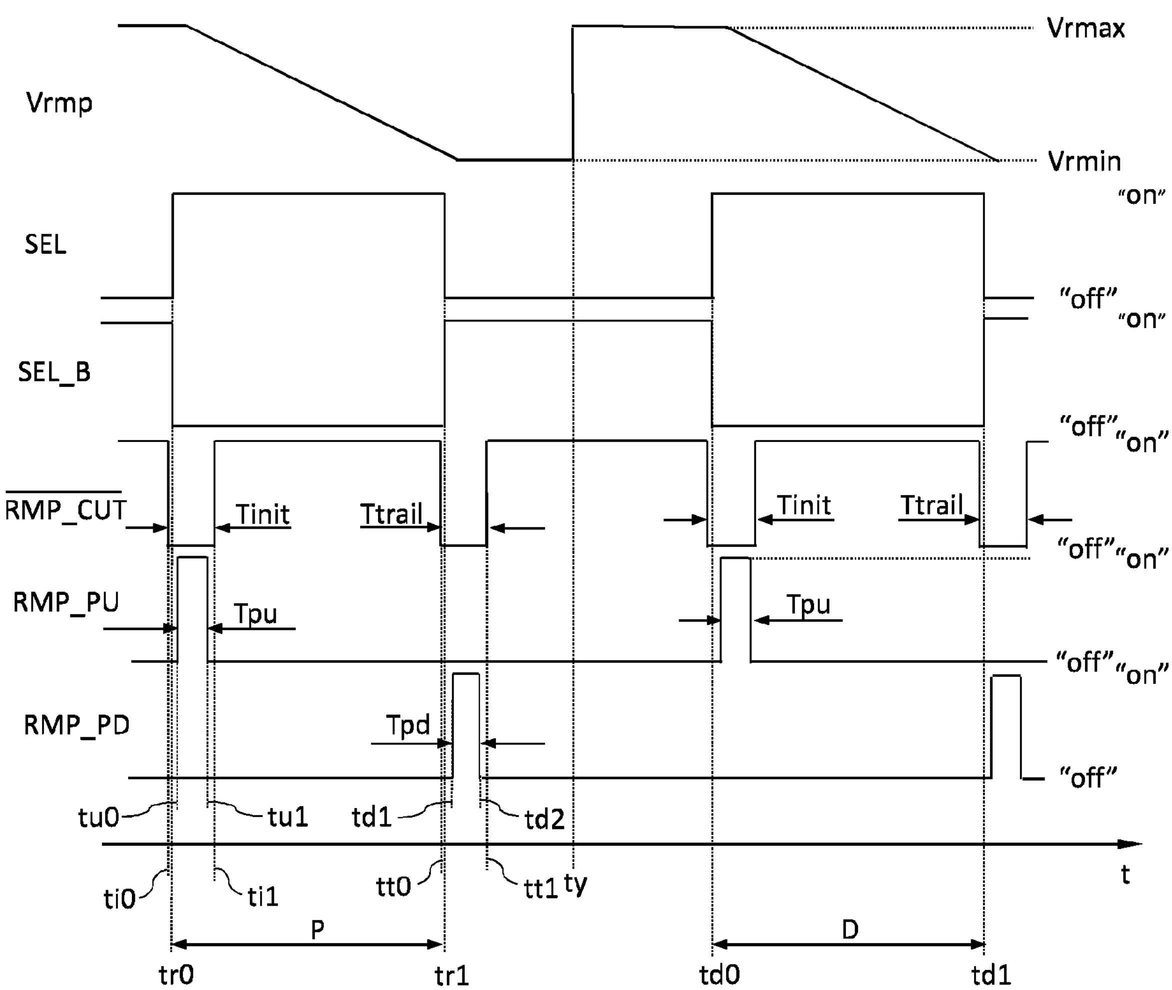

VDD
100
Vfd
103
SEL
109
Vout
VSL
240
SEL_B
GND
VDD
250
RMP_PU
253
210
220
252
Out
Vrmp
RMP_CUT
RMP_PD
254
10
GND

FIG 8



Vrmax
Vrmp
Vrmin
"on"
SEL
"off"
"on"
SEL_B
"off"
"on"
RMP_CUT
Tinit
Ttrail
Tinit
Ttrail
"off"
"on"
RMP_PU
Tpu
Tpu
"off"
"on"
RMP_PD
Tpd
"off"
tu0
tu1
td1
td2
t
ti0
ti1
tt0
tt1
ty
P
D
tr0
tr1
td0
td1

FIG 9

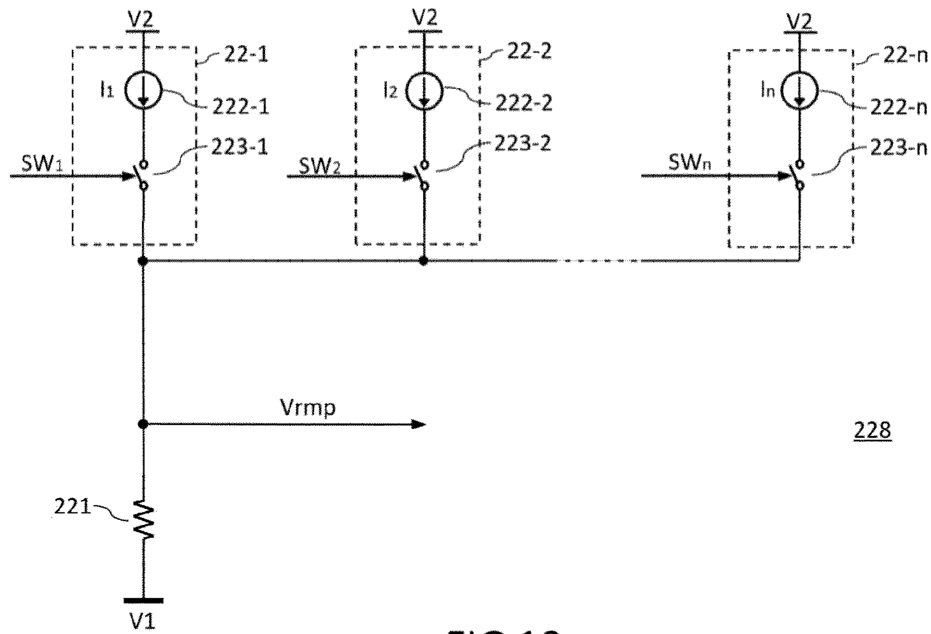

V2
V2
V2
22-1
22-2
22-n
I1
222-1
I2
222-2
In
222-n
SW1
223-1
SW2
223-2
SWn
223-n
Vrmp
228
221
V1

FIG 10

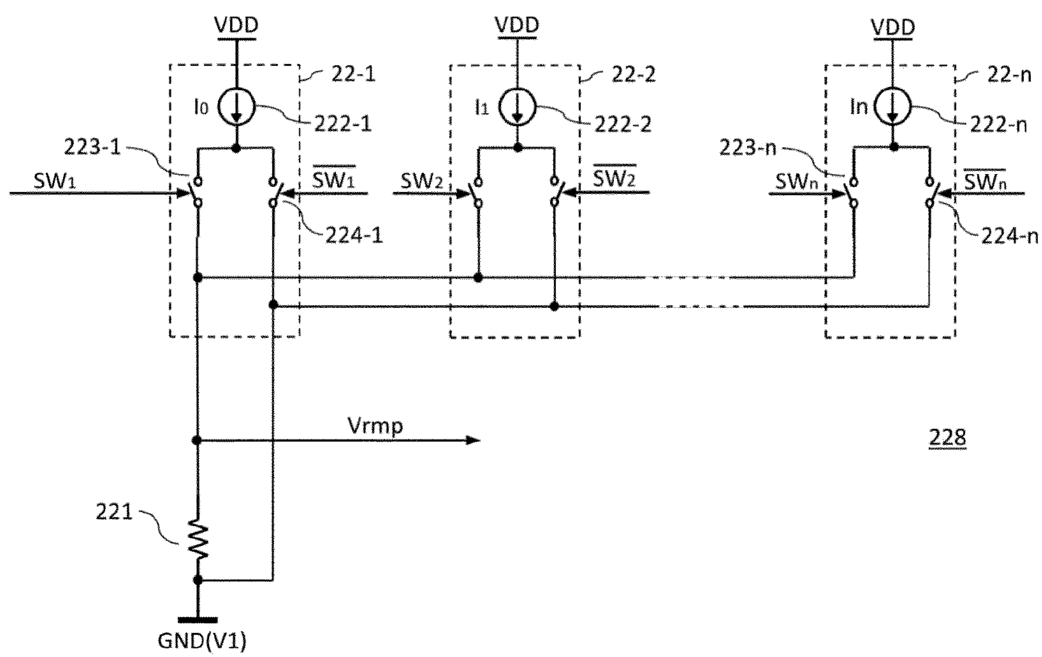

VDD
VDD
VDD
22-1
22-2
22-n
I0
222-1
I1
222-2
In
222-n
223-1
223-n
SW1
SW1
SW2
SW2
SWn
SWn
224-1
224-n
Vrmp
228
221
GND(V1)

FIG 11

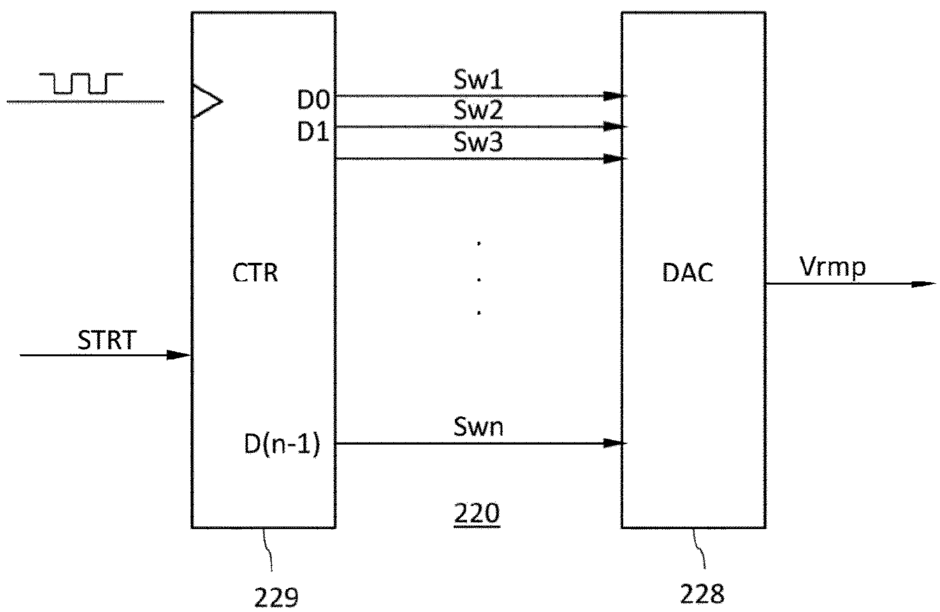


FIG 12

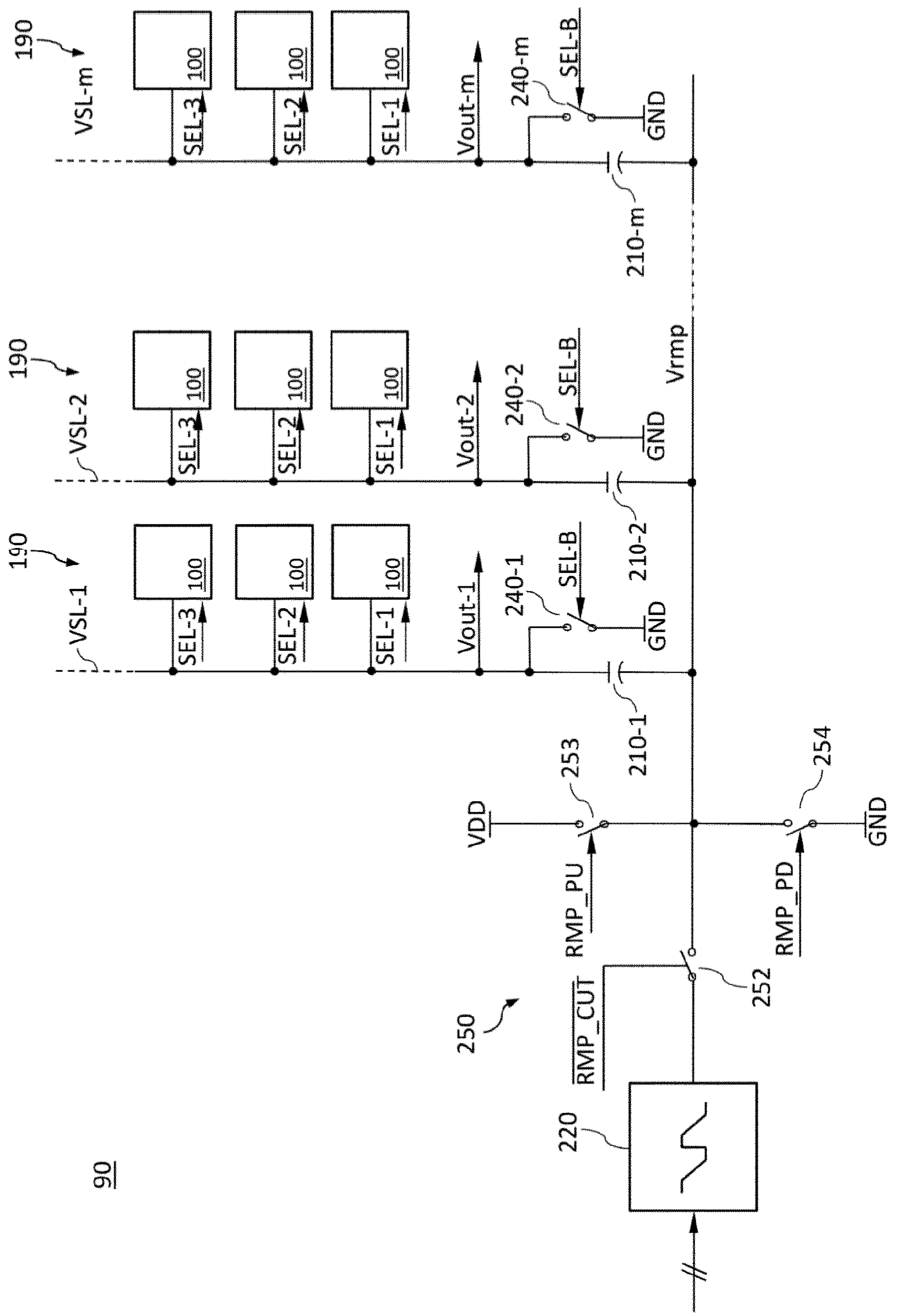

90
190
VSL-1
VSL-2
VSL-m
SEL-1
SEL-2
SEL-3
100
Vout-1
Vout-2
Vout-m
240-1
240-2
240-m
SEL-B
GND
210-1
210-2
210-m
Vrmp
VDD
253
254
RMP_PU
RMP_PD
RMP_CUT
252
250
220

FIG 13

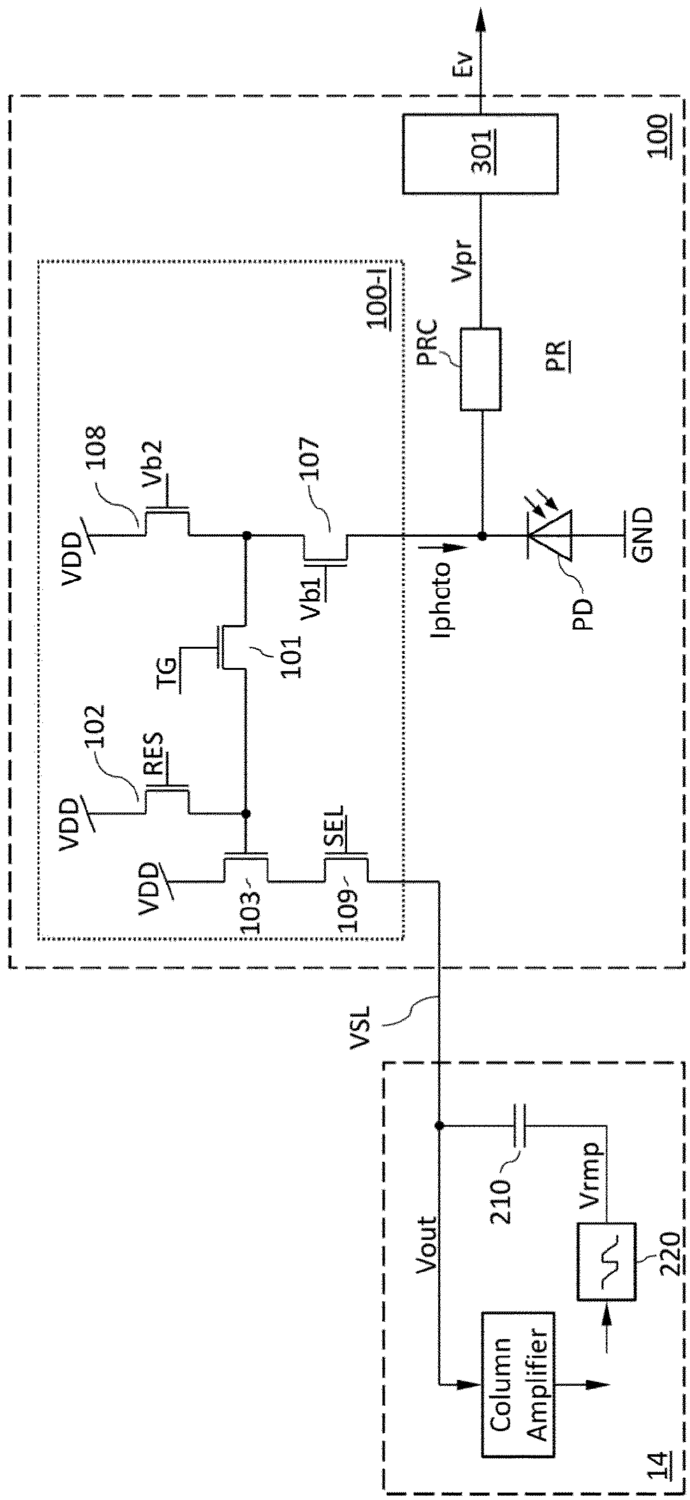

Ev
301
100
100-I
Vpr
PRC
PR
108
Vb2
107
VDD
Vb1
Iphcto
PD
GND
TG
101
102
RES
SEL
103
109
VSL
210
Vrmp
Vout
220
Column Amplifier
14

FIG 14

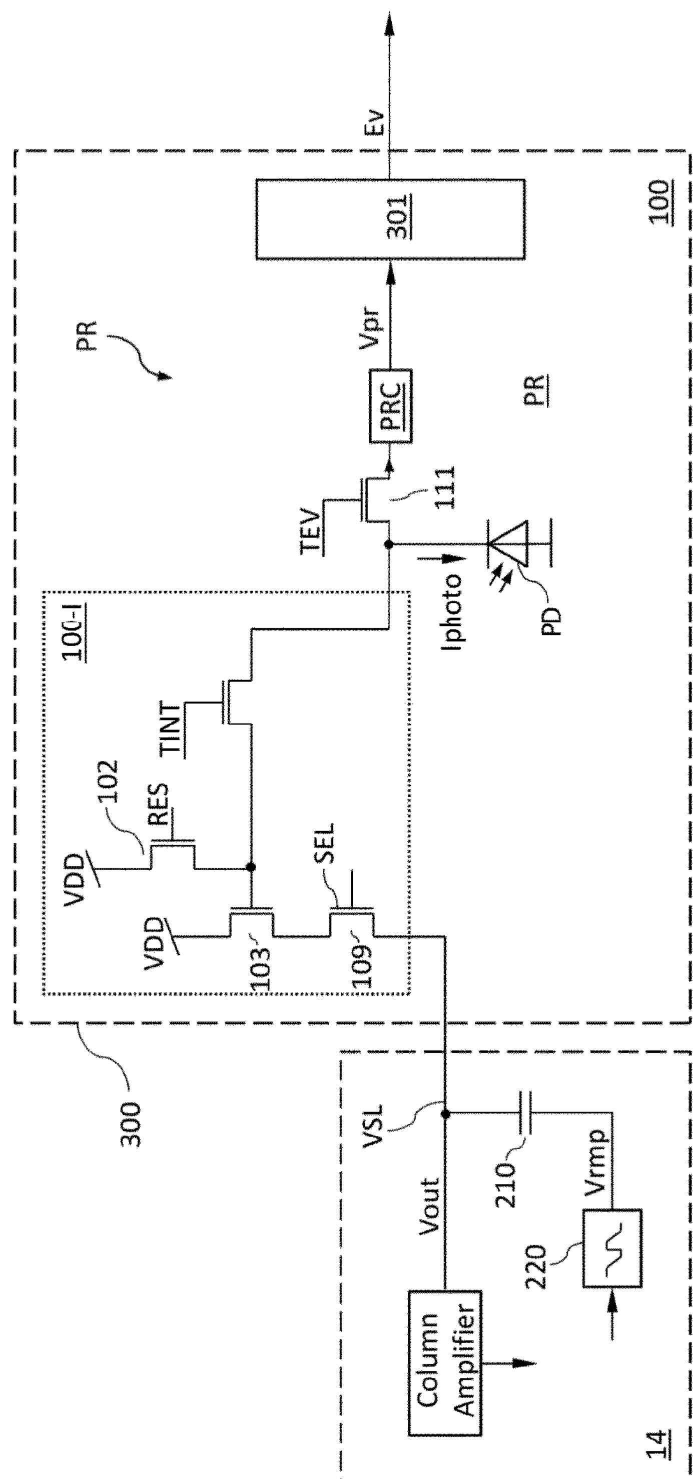

Ev
301
100
Vpr
PR
PRC
PR
TEV
111
Iphoto
PD
100-I
TINT
102
RES
VDD
SEL
VDD
103
109
300
VSL
Vout
210
Vrmp
220
Column Amplifier
14

FIG 15

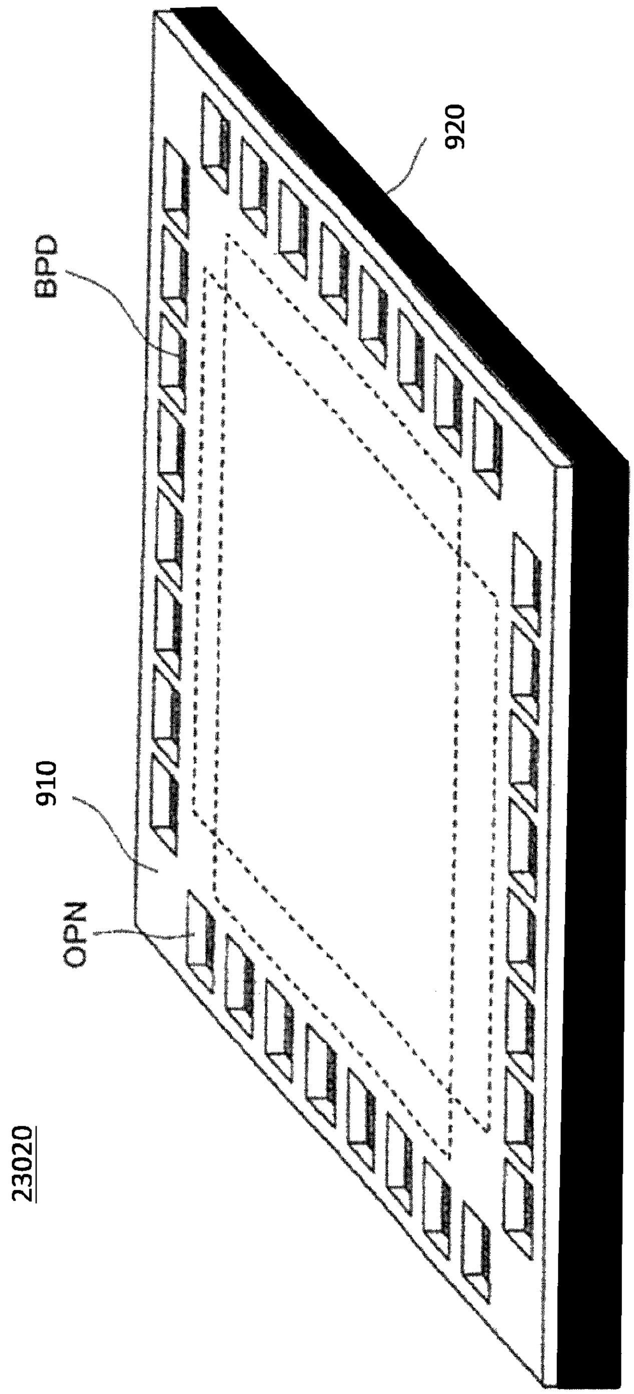


FIG 16

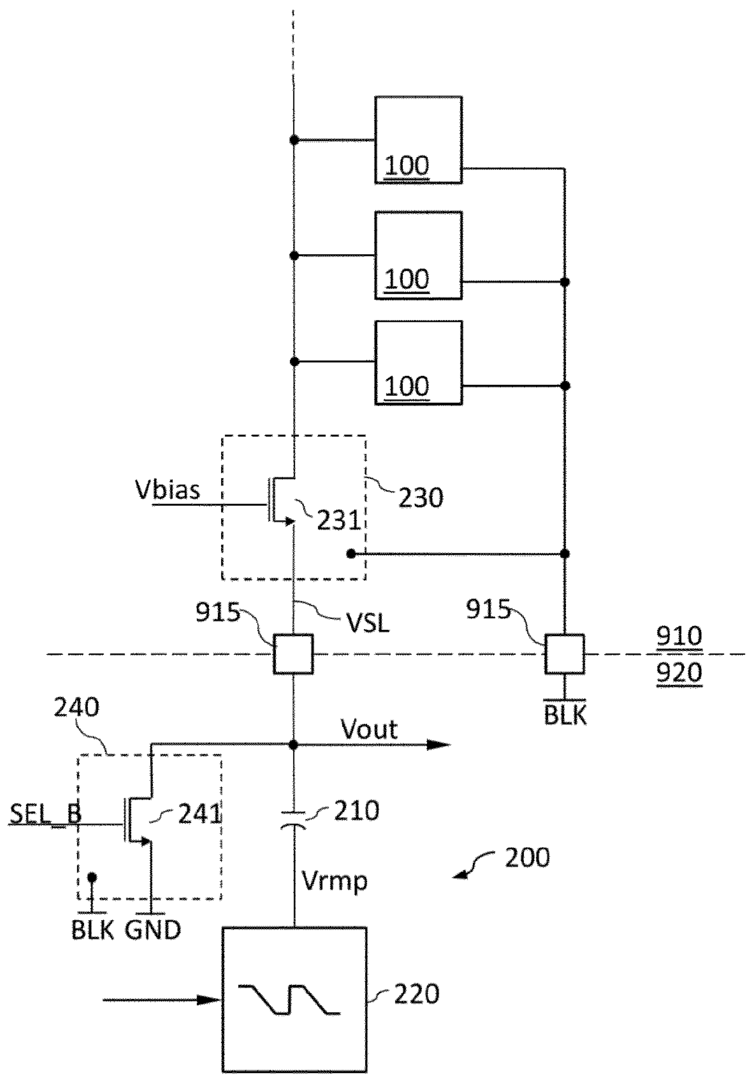

100
100
100
Vbias
231
230
915
VSL
915
910
920
BLK
240
Vout
SEL_B
241
210
200
Vrmp
BLK
GND
220

FIG. 17

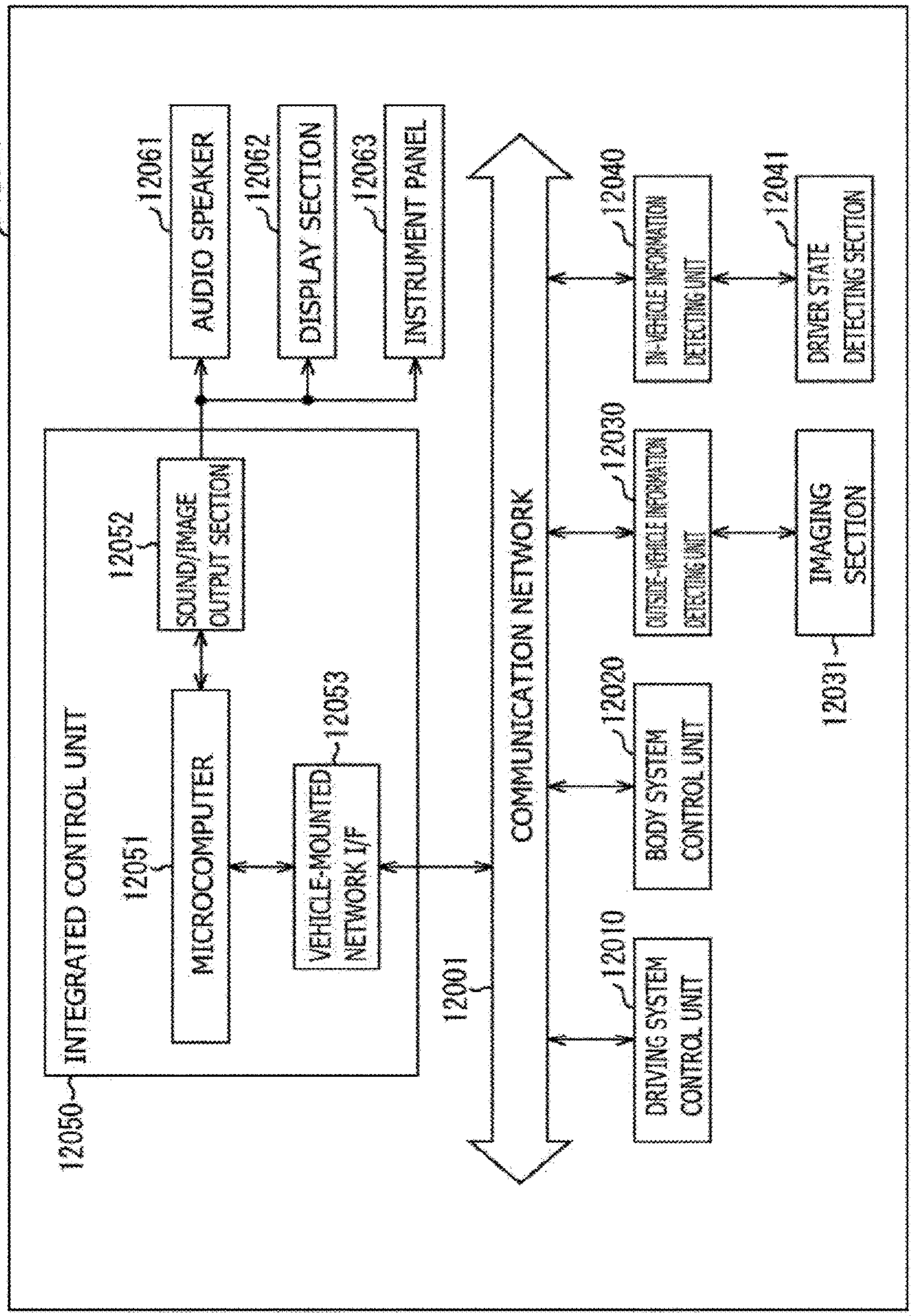

12000
12050
INTEGRATED CONTROL UNIT
12051
MICROCOMPUTER
12052
SOUND/IMAGE OUTPUT SECTION
12053
VEHICLE-MOUNTED NETWORK I/F
12061
AUDIO SPEAKER
12062
DISPLAY SECTION
12063
INSTRUMENT PANEL
12001
COMMUNICATION NETWORK
12010
DRIVING SYSTEM CONTROL UNIT
12020
BODY SYSTEM CONTROL UNIT
12030
OUTSIDE-VEHICLE INFORMATION DETECTING UNIT
12031
IMAGING SECTION
12040
IN-VEHICLE INFORMATION DETECTING UNIT
12041
DRIVER STATE DETECTING SECTION

FIG 18

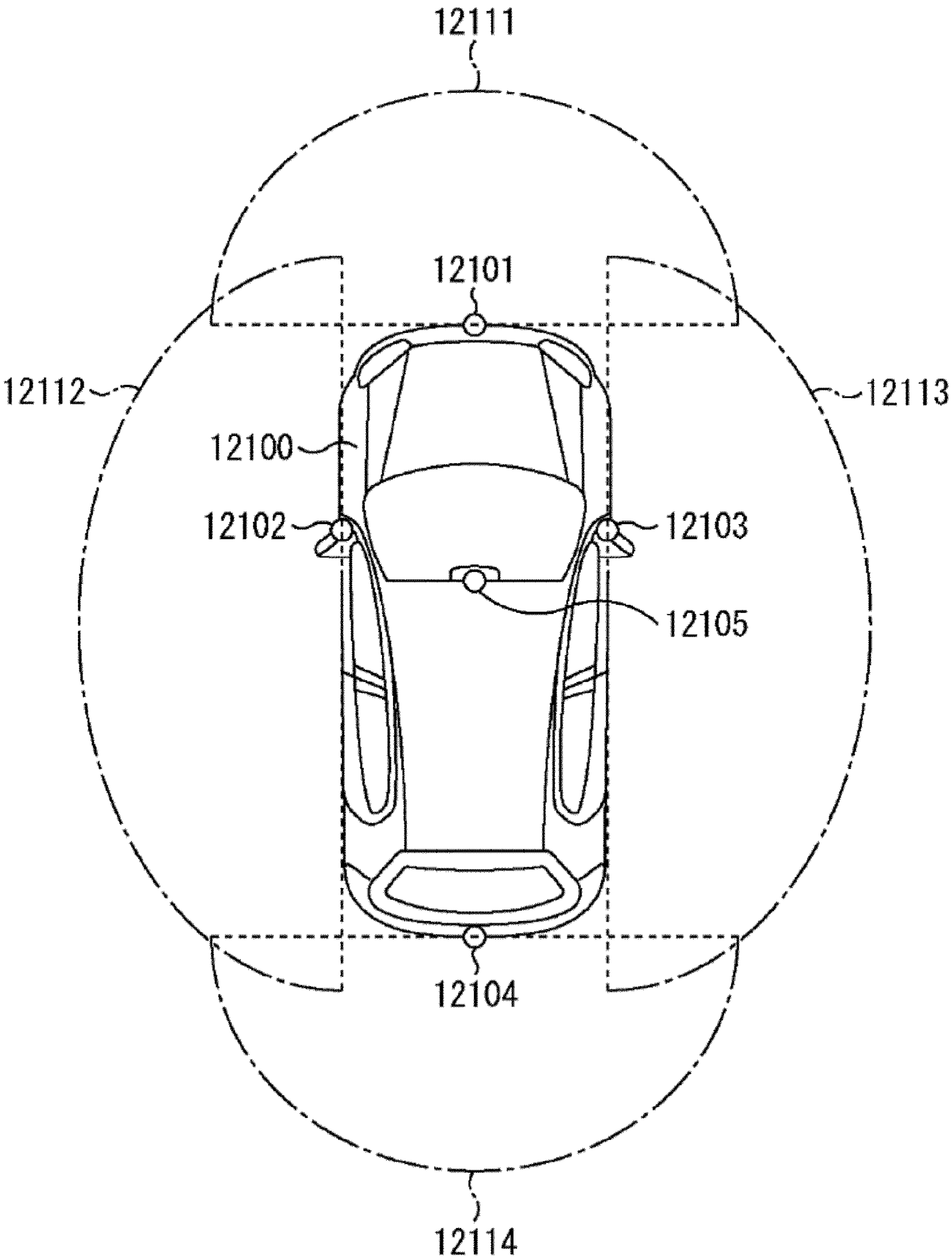

12111
12101
12112
12113
12100
12102
12103
12105
12104
12114

IMAGE SENSOR ARRAY WITH CAPACITIVE CURRENT SOURCE AND SOLID-STATE IMAGING DEVICE COMPRISING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is based on PCT filing PCT/EP2022/057484, filed Mar. 22, 2022, and claims priority from European Patent Application No. 21165244.1, filed Mar. 26, 2021, the entire contents of each are incorporated herein by reference.

The present disclosure relates to an image sensor array including a column signal processing unit and to a solid-state imaging device. In particular, the disclosure relates to the processing of pixel output signals of pixel circuits.

BACKGROUND

Image sensors in solid-state imaging devices include photoelectric conversion elements generating a photocurrent in proportion to the received radiation intensity. A pixel circuit transforms the small photocurrent generated by the photoelectric conversion element into a comparatively large output voltage which a downstream analog-to-digital converter converts into a digital signal. Typically, the pixel circuit includes the amplifier portion of a source follower circuit (common-drain amplifier) that passes the pixel output signal to a data signal line. The data signal line is shared by a plurality of pixel circuits assigned to the same pixel column and the pixel output signals of the pixel column are output individually in a time multiplex regime. A column signal processing unit sequentially receives and processes the pixel output signals.

It is desirable to further improve the dynamic range of image sensor arrays and to reduce noise effects.

SUMMARY

The source follower circuits connected to the same data signal line share a common constant current source as emitter resistor, wherein the constant current source is typically an nFET (n channel field effect transistor) with constant gate bias. For a linear response of the source follower circuits, the nFET is operated in the saturation region. Operation in the saturation region requires that a voltage drop between drain and emitter of the nFET is greater than a minimum voltage, e.g. greater than 300 mV to 400 mV.

The present disclosure mitigates shortcomings of image sensor arrays for intensity read-out attributed to the constant current sources used for pixel read-out.

To this purpose, an image sensor array according to the present disclosure includes a pixel circuit that generates a pixel output signal, wherein an amplitude of the pixel output signal is related to an intensity of detected light. The pixel circuit passes the pixel output signal to a data signal line for a selection period. A current control capacitor supplies a current to the data signal line through a first electrode in the selection period. A ramp generator generates a voltage ramp signal and passes the voltage ramp signal to a second electrode of the current control capacitor in the selection period

A solid-state imaging device according the present disclosure includes a plurality of pixel circuits, wherein each pixel circuit generates a pixel output signal with an amplitude related to an intensity of detected light. Each pixel circuit is assigned to one of a plurality of pixel columns. The solid-state imaging device further includes a plurality of data signal lines and a plurality of current control capacitors. Each data signal line receives the pixel output signals of a plurality of pixel circuits assigned to a same pixel column. A selected one of the pixel circuits passes the pixel output signal to the data signal line for a selection period. Each current control capacitor supplies a current to one of the data signal lines through a first electrode in the selection period. A ramp generator generates a voltage ramp signal and passes the voltage ramp signal to second electrodes of the current control capacitors in the selection period.

BRIEF DESCRIPTION OF THE DRAWINGS

A more complete appreciation of the disclosure and many of the attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings, wherein:

FIG. **1** is a simplified block diagram illustrating an embodiment of a solid-state imaging device according to an embodiment of the present technology.

FIG. **2** is a simplified block diagram illustrating a configuration example of a part of an image sensor array including a current control capacitor with a first electrode connected to a data signal line and with a second electrode receiving a voltage ramp signal according to an embodiment.

FIG. **3** includes time charts for discussing effects of the embodiments on pixel output signals on the data signal line.

FIG. **4** is a simplified circuit diagram illustrating a configuration example of a part of an image sensor array according to an embodiment including an n channel common gate transistor for reducing the capacitive load on the data signal line.

FIG. **5** is a simplified circuit diagram illustrating a configuration example of a part of an image sensor array according to an embodiment including a pre-charge circuit for the data signal line.

FIG. **6** includes time charts for discussing effects of the embodiments on pixel output signals on the data signal line during reset and data phases.

FIG. **7** is a simplified circuit diagram illustrating a configuration example of a part of an image sensor array according to an embodiment including a capacitor control circuit for controlling the current control capacitor.

FIG. **8** includes simplified time charts for discussing effects of the embodiment of FIG. **7**.

FIG. **9** is a simplified circuit diagram illustrating a part of a ramp generator according to an embodiment, for which generating the voltage ramp signal includes digital-to-analog conversion.

FIG. **10** is a simplified circuit diagram illustrating a part of a ramp generator according to another embodiment for which generating the voltage ramp signal includes digital-to-analog conversion.

FIG. **11** is a simplified circuit diagram illustrating a configuration example of a ramp generator including a counter unit.

FIG. **12** is a simplified block diagram illustrating a configuration example of a part of an image sensor array including a ramp generator passing a voltage ramp signal to a plurality of current control capacitors according to a further embodiment.

FIG. **13** is a simplified circuit diagram of a pixel circuit providing simultaneously intensity readout and event detection.

FIG. **14** is a simplified circuit diagram of a pixel circuit switchable between intensity readout and event detection.

FIG. **15** illustrates an overview of a configuration example of a multi-layer solid-state imaging device to which a technology according to the present disclosure may be applied.

FIG. **16** is a schematic circuit diagram of elements of an image sensor array formed on a second chip of a solid-state imaging device with laminated structure according to an embodiment.

FIG. **17** is a block diagram depicting an example of a schematic configuration of a vehicle control system.

FIG. **18** is a diagram of assistance in explaining an example of installation positions of an outside-vehicle information detecting section and an imaging section of the vehicle control system of FIG. **17**.

DETAILED DESCRIPTION

Embodiments for implementing techniques of the present disclosure (also referred to as "embodiments" in the following) will be described below in detail using the drawings. The techniques of the present disclosure are not limited to the described embodiments, and various numerical values and the like in the embodiments are illustrative only. The same elements or elements with the same functions are denoted by the same reference signs. Duplicate descriptions are omitted.

Connected electronic elements may be electrically connected through a direct, permanent low-resistive connection, e.g., through a conductive line. The terms "electrically connected" and "signal-connected" may also include a connection through other electronic elements provided and suitable for permanent and/or temporary signal transmission and/or transmission of energy. For example, electronic elements may be electrically connected or signal-connected through resistors, capacitors, and electronic switches such as transistors or transistor circuits, e.g. MOSFETs, transmission gates, and others.

The load path of a transistor is the controlled path of a transistor. For example, a voltage applied to a gate of a field effect transistor (FET) controls by field effect the current flow through the load path between source and drain.

Though in the following a technology for increasing dynamic range for pixel sensors is described in the context of certain types of active image sensors for intensity readout, the technology may also be used for other types of image sensors.

FIG. **1** illustrates a configuration example of a solid-state imaging device **90** including an image sensor assembly **10** and a signal processing unit **80** according to an embodiment of the present technology.

The image sensor assembly **10** may include a pixel array unit **11**, a row decoder **12**, a pixel driver unit **13**, a column signal processing unit **14**, and a sensor controller **15**.

The pixel array unit **11** includes a plurality of pixel circuits **100**. Each pixel circuit **100** includes a photoelectric conversion element PD and a number of FETs (field effect transistors) for controlling the signal output by the photoelectric conversion element PD. The photoelectric conversion devices PD may be arranged matrix-like in columns and rows. A subset of pixel circuits **100** assigned to the same column of photoelectric conversion devices PD forms a pixel column. The outputs of the pixel circuits **100** of the same pixel column are successively supplied to a data signal line (vertical signal line) VSL.

The row decoder **12** and the pixel driver unit **13** control driving of each pixel circuit **100** disposed in the pixel array unit **11**. In particular, the row decoder **12** may supply a control signal for selecting the pixel circuit **100** or the row of pixel circuits **100** to be driven to the pixel driver unit **13** according to an address signal from the sensor controller **15**. The pixel driver unit **13** may drive the FETs of the selected pixel circuit **100** according to driver timing signals supplied from the sensor controller **15** and the control signal supplied from the row decoder **12**.

The data signal lines VSL pass the output signals of the pixel circuits **100** (pixel output signals Vout) to the column signal processing unit **14**.

For pixel circuits **100** implementing intensity readout, the column signal processing unit **14** may include one or more ADCs (analog-to-digital converters **20**. The column signal processing unit **14** may include as much ADCs **20** as the pixel army unit **11** includes data signal lines VSL. Alternatively, the number of ADCs **20** may be lower than the number of data signal lines VSL and each ADC **20** may be multiplexed between two or more of the data signal lines VSLs. Each ADC **20** performs an analog-to-digital conversion on the pixel output signals successively passed from the pixel column and passes digital pixel data DPXS to the signal processing unit **80**. To this purpose, each ADC **20** may include a comparator **23**, a digital-to-analog converter **22** and a counter **24**.

The sensor controller **15** controls the components of the image sensor assembly **10**. For example, the sensor controller **15** may generate and pass the address to the row decoder **12**, and may generate and pass driving timing signals to the pixel driver unit **13**. In addition, the sensor controller **15** may generate and pass one or more control signals to the column signal processing unit **14**, e.g. to the ADCs **20**.

The pixel circuits **100** may be any active pixel sensors for intensity readout. The illustrated example refers to pixel circuits **100** for intensity readout with one photoelectric conversion element PD and four transistors (FETs) as active elements.

The photoelectric conversion element PD may include or may be composed of, for example, a photodiode. The FETs may include a transfer transistor **101**, a reset transistor **102**, an amplification transistor **103**, and a selection transistor **109**.

The photoelectric conversion element PD photoelectrically converts incident electromagnetic radiation into electric charges. The amount of electric charge generated in the photoelectric conversion element PD corresponds to the intensity of the incident electromagnetic radiation. For example, the photoelectric conversion element PD may include or consist of a photodiode which converts electromagnetic radiation incident on a detection surface into a detector current by means of the photoelectric effect. The electromagnetic radiation may include visible light, infrared radiation and/or ultraviolet radiation. The amplitude of the detector current corresponds to the intensity of the incident electromagnetic radiation, wherein in the intensity range of interest the detector current increases approximately linearly with increasing intensity of the detected electromagnetic radiation.

The transfer transistor **101** is connected between the photoelectric conversion element PD and a floating diffusion region FD. The transfer transistor **101** serves as transfer element for transferring charge from the photoelectric conversion element PD to the floating diffusion region FD. The floating diffusion region FD serves as temporary local charge storage. A transfer signal TG serving as a control signal is supplied to the gate (transfer gate) of the transfer transistor **101** through a transfer control line. Thus, the transfer transistor **101** may transfer electrons photoelectrically converted by the photoelectric conversion element PD to the floating diffusion region FD.

The reset transistor **102** is connected between the floating diffusion region FD and a power supply line to which a positive supply voltage VDD is supplied. A reset signal RES serving as a control signal is supplied to the gate of the reset transistor **102** through a reset control line. Thus, the reset transistor **102** serving as a reset element resets a floating diffusion potential Vfd of the floating diffusion region FD to that of the power supply line supplying the positive supply voltage VDD.

The floating diffusion region FD is connected to the gate of the amplification transistor **103** serving as an amplification element. The floating diffusion region FD functions as the input node of the amplification transistor **103**.

The amplification transistor **103** and the selection transistor **109** are connected in series between the power supply line and the data signal line VSL. Thus, the amplification transistor **103** is connected to the data signal line VSL through the selection transistor **109**.

A selection signal SEL serving as a control signal corresponding to an address signal is supplied to the gate of the selection transistor **109** through a selection control line, and turns on the selection transistor **109**. When the selection transistor **109** is turned on, the amplification transistor **103** amplifies the floating diffusion potential Vfd of the floating diffusion region FD and outputs a voltage corresponding to the floating diffusion potential Vfd to the data signal line VSL. The data signal line VSL passes the pixel output signal Vout from the pixel circuit **100** to the column signal processing unit **14**.

Since the respective gates of the transfer transistor **101**, the reset transistor **102**, and the selection transistor **109** are, for example, connected in units of pixel rows, these operations may be simultaneously performed for each of the pixel circuits **100** of one pixel row.

The data signal line VSL is further connected to a constant current circuit **200** that includes a current control capacitor **210** and a ramp generator **220** controlled by the sensor controller **15**. The constant current circuit **200** may be configured to be effective as switched capacitor current source supplying at least temporarily a constant current to the data signal line VSL.

The amplifier transistor **103** of the pixel circuit **100** and the constant current circuit **200** complement to a source follower circuit passing a pixel output signal Vout derived from the floating diffusion potential Vfd to the column signal processing unit **14** and the ADCs **20**. The ADCs **20** transform the received pixel output signals Vout into digital pixel data DPXS and pass the digital pixel data DPXS to the signal processing unit **80**.

FIG. **2** refers to details of an image sensor array **10** according to the present disclosure.

The image sensor array **10** includes a pixel circuit **100** that generates a pixel output signal Vout with an amplitude related to an intensity of detected light and passes the pixel output signal Vout to a data signal line VSL for a selection period. A current control capacitor **210** supplies a current to the data signal line VSL through a first electrode in the selection period. A ramp generator **220** generates a voltage ramp signal Vrmp and passes the voltage ramp signal Vrmp to a second electrode of the current control capacitor **210** in the selection period.

In particular, the pixel circuit **100** may include an amplification transistor **103** with a load path connected between a positive supply voltage VDD and a pixel output node PON. A floating diffusion region may be connected to a gate of the amplification transistor **103**. A floating diffusion potential Vfd may be passed to the gate of the amplification transistor **103** and controls the amplification transistor **103**.

A selection transistor **109** is connected with a load path between the pixel output node PON and the data signal line VSL. A gate of the selection transistor **109** receives a selection signal SEL. The selection signal SEL is active ("on") and turns on the selection transistor **109** for a selection period. Outside the selection period, the selection signal SEL is inactive ("off") and turns off the selection transistor **109**.

The substrate bulk for the amplification transistor **103** and the selection transistor **109** is connected to a bulk potential BLK. The bulk potential BLK may be connected to a voltage reference potential GND or to a negative potential.

A first electrode of the current control capacitor **210** (first control capacitor electrode) is signal-connected to the data pixel circuit **100** through the data signal line VSL. The first control capacitor electrode may be directly connected to the pixel circuit **100**. Alternatively, further elements or circuits facilitating at least temporary signal transmission such as FETs, switches and/or resistors may be electrically connected between the pixel circuit **100** and the first control capacitor electrode and signal-connect the pixel circuit **100** with the first control capacitor electrode in at least a part of the selection period.

The ramp generator **220** generates a voltage ramp signal Vrmp in response to one or more control signals received from the sensor controller and outputs the voltage ramp signal Vrmp at a ramp generator output Out. The ramp generator output Out is signal-connected to the second electrode of the current control capacitor **210** (second control capacitor electrode). The ramp generator output Out and the second control capacitor electrode may be directly connected. Alternatively, further elements or circuits allowing at least temporary signal transmission such as FETs, switches and/or resistors may be connected between the ramp generator output Out and the second control capacitor electrode and connect the ramp generator output Out with the second control capacitor electrode for at least a part of the selection period.

When a first capacitor electrode of a capacitor with a capacitance C is kept at constant voltage level while a voltage ramp with a constant slope ratio Vrmp/Trmp is applied to a second capacitor electrode, the first capacitor electrode supplies a constant current $Ic=(Vrmp*C)/Trmp$ to the circuit connected to the first capacitor electrode.

In particular, during the selection period the amplification transistor **103** forces the voltage level of the pixel output signal Vout on the data signal line VSL to a level that is a function of the floating diffusion potential Vfd provided that a current source supplies a constant current to the data signal line VSL. The current control capacitor **210** supplies the required constant current to the data signal line VSL through the first control capacitor electrode provided that simultaneously a sufficiently steep voltage ramp is supplied to the second control capacitor electrode.

The constant current circuit **200** and in particular the current control capacitor **210** and the amplification transistor **103** complement each other to a source follower circuit, wherein the current control capacitor **210** is effective as the constant current source for the output current.

The time charts in FIG. **3** show the effect of the constant current circuit **200** of FIG. **2**. The upper time chart (first time chart) shows a time response of the floating diffusion potential Vfd. The second time chart shows the voltage ramp signal Vrmp. The third time chart shows the resulting pixel output signal Vout on the data signal line VSL. The pixel circuit may be continuously selected from t=tx0 to t=tx4.

In general, one pixel readout cycle may include a preset phase P and a data phase D. A voltage level of the pixel output signal Vout at the end of the preset phase P may indicate a current pixel offset voltage obtained as output signal of a pixel circuit with unilluminated photoelectric conversion element PD ("dark pixel"). A voltage level of the pixel output signal Vout at the end of the data phase D is a measure for incident light on the illuminated photoelectric conversion element PD. For CDS (correlated double sampling), the kTC noise may be canceled by subtracting the current pixel offset voltage obtained in the preset phase P from the pixel output signal Vout at the end of the data phase D. A settling time after which the voltage level of the pixel output signals Vout is stable after transitions to the preset phase P and to the data phase D determines the readout speed.

At t=tx0 the floating diffusion region is connected to a constant reset potential. For example, the reset transistor **102** shown in FIG. **1** may connect the floating diffusion region to the high supply voltage VDD. Accordingly, the floating diffusion potential Vfd is equal to a reset potential. The reset potential may be about 1.4V.

Prior to tx0, at t=tx0, or shortly later the voltage ramp signal Vrmp supplied to the second control capacitor electrode starts to drop at constant rate. The slope ratio of the voltage ramp is equal to 1V/μs and the capacitance C of the current control capacitor **210** in FIG. **2** is 5 pF such that the current control capacitor **210** supplies a constant current of 5 μA to the data signal line VSL.

At t=tx1, after a settling time of 200 ns to 300 ns, the pixel output signal Vout reaches a constant voltage level that depends on the reset potential of the floating diffusion region.

At t=tx2 the floating diffusion region is disconnected from the constant reset potential and connected to the photoelectric conversion element of the pixel circuit. The floating diffusion potential Vfd adapts according to the intensity of radiation detected by the photoelectric conversion element and reaches the corresponding final constant voltage level at t=tx3. The different floating diffusion potentials Vfd1, . . . , Vfd10 correspond to different illumination conditions, wherein the intensity of detected radiation increases from Vfd1 to Vfd10 and wherein Vfd1 is the floating diffusion potential of an unexposed pixel (dark pixel) and Vfd10 is the floating diffusion potential of a fully exposed pixel.

From t=tx0 to t=tx4 the constant current circuit **200** steadily supplies a constant current of 5 μA such that the pixel output signal Vout follows the floating diffusion potential Vfd accordingly and reaches constant voltage levels for the different illumination conditions at t=tx3.

The lower limit of the output voltage swing for the pixel output signal Vout (pixel output voltage swing) is given by the bulk potential BLK to which the substrate bulks of the amplification transistor **103** and the selection transistor **109** are connected. In case the bulk potential BLK corresponds to the voltage reference potential GND (0V), the lower limit of the pixel output voltage swing is equal to 0V.

In comparison with a conventional constant current source based on an nFET with constant gate bias, the constant current circuit **200** of FIG. **2** provides a capacitor current source that increases the voltage swing of the pixel output voltage Vout by 300 mV to 400 mV, in other words, by about 40%. The increased pixel output voltage swing in turn facilitates a higher voltage swing for the floating diffusion potential Vfd and eventually a higher dynamic range. The pixel output voltage swing may be further increased by supplying a negative potential as bulk potential BLK.

Further in comparison with a conventional constant current source based on an nFET with constant gate bias, the capacitor current source eliminates the 1/f and RTS (random telegraph signal) noise inherently introduced by the operation of FETs.

FIG. **4** shows a capacitance control circuit **230** connected between the data signal line VSL and the current control capacitor **210**. The capacitance control circuit **230** includes an n channel common gate transistor for reducing a capacitive load effective for the pixel output signal Vout.

In particular, if the capacitance Cramp of the current control capacitor **210** is large compared to the capacitance of the data signal line VSL, the capacitance control circuit **230** may be efficient to reduce the settling time, i.e. the time elapsed until the pixel output signal Vout reaches a specified error band around the final voltage level after a voltage transition. The capacitance control circuit **230** may include any circuit that reduces the capacitive load on the data signal line VSL.

For example, the capacitance control circuit **230** may include at least one FET **231**, wherein a load path of the FET **231** is connected between the pixel circuit **100** and the first electrode of the current control capacitor **210**.

In the illustrated embodiment, the capacitance control circuit **230** includes one nFET **231** with constant gate bias Vbias.

FIG. **5** shows a pre-charge circuit **240** that pre-charges the data signal line VSL in a pre-charge period outside the selection period.

The data signal line VSL may be pre-charged to a constant voltage lower than the positive supply voltage VDD, for example to the voltage reference potential GND. The pre-charge circuit **240** may be active in pre-charge periods prior to settling phases, e.g. prior to each selection period. The pre-charge circuit **240** is inactive for the selection periods.

By pre-charging the data signal line VSL with the voltage reference potential GND, the settling of the pixel output signal Vout concerns a transition from a low voltage to a high voltage. The transition from the low voltage to the high voltage is governed by the source follower configuration including the amplification transistor such that the output load is driven by the comparatively high source follower current. Otherwise, if the data signal line VSL is pre-charged with high potential, the settling of the pixel output signal Vout concerns transitions form a high voltage to a low voltage, wherein the output load is driven by the comparatively low current supplied by the constant current circuit **200**.

Therefore, in particular if the capacitance Cramp of the current control capacitor **210** is small compared to the capacitance of the data signal line VSL, by pre-charging the data signal line VSL prior to the preset phase P and prior to the data phase D with the voltage reference potential GND, the pre-charge circuit **240** may efficiently reduce, for a given constant current supplied by the current control capacitor **210**, the settling time, i.e. the time elapsed until the pixel output signal Vout reaches a specified error band around the final voltage level after a voltage transition.

The pre-charge circuit **240** may include a first switching element **241** that passes the voltage reference potential GND to the data signal line VSL in the pre-charge period. A pre-charge control signal SEL_B may control the first switching element **241**.

The pre-charge circuit **240** may be configured to start each pre-charge period with end of a preceding selection period and to end each pre-charge period with start of a following selection period.

The selection periods at the data signal line VSL may concern different pixel circuits which are assigned to the same data signal line VSL and selected with different select signals.

The time charts in FIG. **6** illustrate the effect of the pre-charge circuit **240** of FIG. **5**. The upper time chart (first time chart) shows a time response of the floating diffusion potential Vfd. The second time chart shows a time response of the selection signal SEL. The third time chart shows a time response of the pre-charge signal SEL_B. For simplicity, one single pixel circuit per data signal line is considered in this example, wherein the pre-charge control signal SEL_B may be approximated by the inverted selection signal SEL. The fourth time chart shows the voltage ramp signal Vrmp. The fifth time chart shows the resulting pixel output signal Vout on the data signal line VSL.

Active periods of the selection signal SEL define selection periods. A first selection period defines a preset phase P between t=tr0 and t=tr1. A second selection period defines a data phase D between t=td0 and t=td1.

In the preset phase P, the floating diffusion region is connected to a constant reset potential, e.g., to the high supply voltage VDD. Starting with begin of the preset phase P, the voltage ramp signal Vrmp drops at constant rate. The slope ratio of the voltage ramp may be set such that within the preset phase P the voltage ramp signal drops form a maximum ramp voltage Vrmax to a minimum ramp voltage Vrmin. At t=tx the pixel output signal Vout reaches a constant voltage level that depends on the reset potential of the floating diffusion region. The slope ratio of the voltage ramp is equal to 3V/μs and the capacitance C of the current control capacitor **210** in FIG. **5** is 1 pF such that the current control capacitor **210** supplies a constant current of 3 μA.

At t=tr1 the preset phase P ends. The selection signal SEL becomes inactive and disconnects the pixel circuit **100** from the data signal line VSL. The pre-charge control signal SEL_B turns on the first switching element **241** and connects the data signal line VSL to the voltage reference potential GND, wherein the data signal line VSL is pre-charged with the voltage reference potential GND. At t=ty the ramp voltage signal Vrmp returns to the maximum ramp voltage Vrmax.

Starting with t=tx2 floating diffusion potential Vfd adapts according to the intensity of radiation detected by the photoelectric conversion element and reaches the corresponding final constant level at t=tx3. Different floating diffusion potentials Vfd1, . . . , Vfd11 result from different illumination conditions, wherein the intensity of detected radiation increases from Vfd1 to Vfd11.

Starting with begin of the data phase D at t=td0, the voltage ramp signal Vrmp again drops at the same rate as in the preset phase P.

During both the preset phase P and the data phase D the constant current circuit **200** steadily supplies a constant current of 3 μA such that the pixel output signal Vout follows the floating diffusion potential Vfd accordingly and reaches constant voltage levels for the different illumination conditions at t=tx3.

FIG. **7** shows a part of an image sensor array **10** including a capacitor control circuit **250** that decouples the second electrode of the current control capacitor **210** from an output of the ramp generator **220** in an initial phase at a start of the selection period and/or in a tail phase at an end of the selection period.

The initial phase may start shorty prior to, simultaneously with of shortly after the start of the selection period and ends within the selection period, e.g. within the first quarter of the selection period. The trail phase may start shortly prior to, simultaneously with of shortly after the end of the selection period and ends before the next selection period.

The capacitor control circuit **250** may reduce or eliminate detrimental interaction between components of the constant current circuit **200** and other circuits connected to the data signal line VSL. For example, the capacitor control circuit **250** may be adapted to reduce effects caused by turning on and/or off the selection transistor **109** on the ramp generator **220**.

In particular, the capacitor control circuit **250** may include a second switching element **252** configured to disconnect the second electrode of the current control capacitor **210** from the output Out of the ramp generator **220** for the initial phase of the selection period and/or for the tail phase. A first ramp control signal RMP_CUT may control the second switching element **252**.

In particular, the first ramp control signal RMP_CUT may be active during the switching transitions of the selection transistor **109** such that the output Out of the ramp generator **220** is disconnected from the current control capacitor **210** during the switching transitions and current spikes on the ramp generator **200** can be reduced. In particular, the ramp generator can be protected from receiving current spikes, in particular from current spikes with transition currents higher than the maximum ramp generator output current.

The second switching element **252** may be an FET with a load path connected between the output Out of the ramp generator **220** and the second electrode of the current control capacitor **210**. The first ramp control signal RMP_CUT may be passed to the gate of the FET.

The capacitor control circuit **250** may include a third switching element **253** that passes a first constant voltage to the second electrode of the current control capacitor **210** in the initial phase of the selection period. The third switching **253** element pushes the second electrode of the current control capacitor **210** to an appropriate constant potential for the time the current control capacitor **210** is disconnected from the ramp generator **220**. The first constant voltage may be a high potential, e.g. the positive supply voltage VDD.

A second ramp control signal RMP_PU controls the third switching element **253**. The third switching element **253** may be an FET with a load path connected between the positive supply voltage VDD and the second electrode of the current control capacitor **210**. The second ramp control signal RMP_PU may be passed to the gate of the FET.

The capacitor control circuit **250** may include a fourth switching element **254** configured to pass a second constant voltage to the second electrode of the current control capacitor **210** in the trail phase at the end of the selection period. The fourth switching element **254** element pulls the second electrode of the current control capacitor **210** to an appropriate constant potential for the time the current control capacitor **210** is disconnected from the ramp generator **220**.

The second constant voltage may be a low potential, e.g. the voltage reference potential GND.

A third ramp control signal RMP_PD controls the fourth switching element **254**. The fourth switching element **254** may be an FET with a load path connected between the voltage reference potential GND and the second electrode of the current control capacitor **210**. The third ramp control signal RMP_PD may be passed to the gate of the FET.

FIG. **8** shows the time charts for a voltage ramp signal Vrmp, the selection signal SEL and the pre-charge control signal SEL_B as described with reference to FIG. **6**, and for the first ramp control signal RMP_CUT, the second ramp control signal RMP_PU, and the third ramp control signal RMP_PD.

The first ramp control signal $\overline{\text{RMP\_CUT}}$ is inactive and switches off the second switching element **252** for initial phases Tinit and for trail phases Ttrail. The initial phases Tinit start at t=ti0 prior to begin of the selection periods P, D at t=tr0 and t=td0 and end at t=ti1 shortly after begin of the selection periods P, D. The trail phases Ttrail start at t=tt0 prior to the end of the selection periods P, D at t=tr1 and t=td1 and end at t=tt1 after the end of the selection periods P, D and prior to the next selection period. Outside the initial phases Tinit and the trail phases Ttrail, the first ramp control signal $\overline{\text{RMP\_CUT}}$ is active and the second switching element **252** is on.

The second ramp control signal RMP_PU is active and switches on the third switching element **253** for pull-up phases Tpu. Each pull-up phase Tpu starts at t=tu0 with or after begin of an initial phase Tinit and ends at t=tu1 shortly prior to or with the end of the initial phase Tinit. Outside the pull-up phases Tpu, the second ramp control signal RMP_PU is inactive and the third switching element **253** is off.

The third ramp control signal RMP_PD is active and switches on the fourth switching element **254** for pull-down phases Tpd. Each pull-down phase Tpd starts at t=tp0 with or after begin of a trail phase Ttrail and ends at t=tp1 shortly prior to or with the end of the trail phase Ttrail. Outside the pull-down phases Tpd, the fourth ramp control signal RMP_PD is inactive and the fourth switching element **254** is off.

The ramp generator **220** may be any circuit that creates or approximates a linear rising or falling voltage with respect to time. For example, the ramp generator **220** may include an analog ramp generator like a bootstrap ramp generator using an operational amplifier in a voltage follower configuration. According to another example, the ramp generator includes a DAC (digital-to-analog converter stage) generating a stair step voltage ramp.

FIG. **9** shows a ramp generator **220** that includes a DAC stage **228** with a plurality of switchable current supply cells **22-1**, . . . , **22-***n* connected in parallel and an output resistor **221**. Each current supply cell **22-1**, . . . , **22-***n* includes a cell current source **222-1**, . . . , **222-***n* and a primary switching element **223-1**, . . . , **223-***n* connected in series. The output resistor **221** is connected in series between the parallel connected current supply cells **22-1**, . . . , **22-***n* and a first constant voltage V**1**. The parallel connected current supply cells **22-1**, . . . , **22-***n* are connected in series between a second constant voltage V**2** and the output resistor **221**.

When the primary switching element **223-***x* of a switchable current supply cell **22-***x* is on, the respective cell current source **222-***x* induces a current flow between the first constant voltage V**1** and the second constant voltage V**2** through the output resistor **221**.

The current supply cells **22-1**, . . . , **22-***n* may include FETs with constant gate bias, may be essentially identical and may supply the same current. The primary switching elements **223-1**, . . . , **223-***n* may be FETs. Switch control signals Sw1, . . . , Swn control the primary switching elements **223-1**, . . . , **223-***n* and turn on selected ones. The switch control signals Sw1, Swn control the current supply cells **22-1**, . . . , **22-***n* in a way that the sum current through the output resistor **221** continuously increases or decreases with time in steps at a step height corresponding to the voltage drop one single of the cell current sources **222-1**, . . . , **222-***n* generates at the output resistor **221**.

In the DAC stage **228** illustrated in FIG. **10** each switchable current cell **22-1**, . . . , **22-***n* further includes a secondary switching element **224-1**, . . . , **224-***n* connected in series between the cell current source **222-1**, . . . , **222-***n* and the first constant voltage V**1**. The inverted switch control signals $\overline{\text{Sw1}}$, . . . , $\overline{\text{Swn}}$ control the secondary switching elements **224-1**, . . . , **224-***n* and turn on selected ones such that during operation each cell current source **222-1**, . . . , **222-***n* supplies the same current at any time and the total current consumption remains constant.

The illustrated DAC stage **228** refers to a ground-based type DAC stage, wherein the first constant voltage V**1** is equal to the voltage reference potential GND and the second constant voltage V**2** is equal to the positive supply voltage VDD. According to a DAC stage of the supply-based type, the switchable current cells **22-1**, . . . , **22-***n* may be connected to the voltage reference potential GND and the output resistor **221** may be connected to the positive supply voltage VDD.

FIG. **11** shows a ramp generator **220** that includes a counter **229** controlling the switchable current supply cells **22-1**, . . . , **22-***n* of any of the DAC stages **228** in FIG. **9** or FIG. **10**. The counter **229** may be a binary counter, increasing a digital count value with each rising or trailing edge of a clock signal and outputting the current digital count value in parallel at data outputs D0, . . . , Dn−1. The data outputs D0, . . . , Dn−1 supply the switch control signals Sw1, . . . , Swn.

FIG. **12** shows a portion of a solid-state imaging device **90**. The solid-state imaging device **90** includes a plurality of pixel circuits **100**, wherein each pixel circuit **100** is configured to generate a pixel output signal with an amplitude related to an intensity of detected light, and wherein each pixel circuit **100** is assigned to one of a plurality of pixel columns **190**.

The solid-state imaging device **90** further includes a plurality of data signal lines VSL-1, . . . , VSL-m, wherein each data signal line VSL-1, . . . , VSL-m is configured to receive the pixel output signals of a plurality of pixel circuits **100** assigned to a same pixel column **190**, and wherein to each data signal line VSL-1, . . . , VSL-m a selected pixel circuit **100** passes the pixel output signal Vout for a selection period;

The solid-state imaging device **90** further includes a plurality of current control capacitor s **210-1**, . . . , **210-***m*, wherein each current control capacitor **210-1**, . . . , **210-***m* is configured to supply a current to one of the data signal lines VSL-1, . . . , VSL-m through a first electrode in the selection periods.

A ramp generator **220** generates a voltage ramp signal and passes the voltage ramp signal to second electrodes of the current control capacitors **210-1**, . . . , **210-***m* in the selection periods.

A row selection signal SEL_**1**, SEL_**2**, . . . , simultaneously selects the pixel circuits **100** of the same pixel row, wherein in each pixel column **190** the pixel circuit **100** assigned to the selected pixel row is selected.

Pre-charge circuits **240-1**, . . . , **240-***m* pre-charge the data signal lines VSL-1, VSL-m in pre-charge periods outside the selection periods defined by the active row selection signals SEL_1, SEL_2, ... For example, a pre-charge signal SEL_B controlling the pre-charge circuits **240-1**, . . . , **240-***m* may be or may approximate the inverted of the ORed row selection signals SEL_1, SEL_2, ...

A capacitor control circuit **250** may decouple the second electrodes of the current control capacitors **210-1**, . . . , **210-***m* from an output of the ramp generator **220** in initial phases at start of the selection periods and/or in tail phases at end of the row selection periods as described with reference to FIGS. **7** and **8**.

FIG. **13** refers to a pixel circuit **100** including an intensity readout circuit **100**-I and a photoreceptor module PR for event detection, wherein the intensity readout circuit **100**-I and the photoreceptor module PR share a common photoelectric conversion element PD. The photoreceptor module PR includes a photoreceptor circuit PRC that converts the photocurrent Iphoto into a photoreceptor signal Vpr, wherein a voltage of the photoreceptor signal Vpr is a function of the photocurrent Iphoto, and wherein in the range of interest the voltage of the photoreceptor signal Vpr increases with increasing photocurrent Iphoto. The photoreceptor circuit PRC may include a logarithmic amplifier. An event detector circuit **301** receives the photoreceptor signal Vpr and generates an event detection signal Ev when a change of the voltage level of the photoreceptor signal Vpr exceeds a predetermined threshold.

The intensity readout circuit **100**-I includes an n-channel anti-blooming transistor **108** and an n-channel decoupling transistor **107** which are electrically connected in series between the high supply voltage VDD and the photoelectric conversion element PD. The anti-blooming transistor **108** and the decoupling transistor **107** may be controlled by fixed bias voltages Vb2, Vb1 applied to the gates. Additional elements, e.g. a controlled path of a feedback portion of the photoreceptor circuit PRC may be electrically connected in series between the decoupling transistor **107** and the photoelectric conversion element PD.

Decoupling transistor **107** may basically decouple the photoreceptor circuit PRC from voltage transients at the center node between the decoupling transistor **107** and the anti-blooming transistor **108**. The anti-blooming transistor **108** may ensure that the voltage at the center node between the decoupling transistor **107** and the anti-blooming transistor **108** does not fall below a certain level given by the difference between the bias voltage Vb2 at the gate of the anti-blooming transistor **108** and the threshold voltage of the anti-blooming transistor **108** in order to ensure proper operation of the photoreceptor circuit PRC.

The source of the n-channel transfer transistor **101** is electrically connected to the center node between the decoupling transistor **107** and the anti-blooming transistor **108**. For the further components of the intensity readout circuit **100**-I, reference is made to the description of the pixel circuit **100** in FIG. **1**.

Alternative embodiments of the intensity readout circuit **100**-I may be realized without transfer transistor **101**, wherein the reset transistor **102** may replace the anti-blooming transistor **108**, and wherein the source of the reset transistor **102** is directly connected to the gate of the amplifier transistor **103**.

In the photoreceptor circuit block of FIG. **13**, the intensity detection circuit **100**-I and the photoreceptor circuit PRC for event detection are electrically connected in series with respect to the photocurrent Iphoto, wherein evaluation of intensity and detection of events may be performed substantially contemporaneously.

The pixel circuit **100** in FIG. **14** includes a first mode selector **111** and a second mode selector **112**. The first mode selector **111** is connected between the cathode of the photoelectric conversion element PD and a photoreceptor circuit PRC. The second mode selector **112** is connected between the cathode of the photoelectric conversion element PD and the amplifier transistor **103** of an intensity readout circuit **100**-I. A first mode selector signal TEV controls the first mode selector **111**. A second mode selector signal TINT controls the second mode selector **112**.

The first and second mode selectors **111**, **112** electrically connect the photoelectric conversion element PD with the photoreceptor circuit PRC in a first operating state and with the intensity readout circuit **100**-I in a second operating state. In addition, the first and second mode selectors **111**, **112** may disconnect the photoelectric conversion element PD from the intensity readout circuit **100**-I in the first operating state and may disconnect the photoelectric conversion element PD from the photoreceptor circuit PRC in the second operating state. The first and second mode selectors **111**, **112** may be electronic switches, for example FETs or transfer gates.

FIG. **15** is a perspective view showing an example of a laminated structure of a solid-state imaging device **23020** with a plurality of pixels arranged matrix-like in array form. Each pixel includes at least one photoelectric conversion element.

The solid-state imaging device **23020** has the laminated structure of a first chip (upper chip) **910** and a second chip (lower chip) **920**.

The laminated first and second chips **910**, **920** may be electrically connected to each other through TC(S)Vs (Through Contact (Silicon) Vias) formed in the first chip **910**.

The solid-state imaging device **23020** may be formed to have the laminated structure in such a manner that the first and second chips **910** and **920** are bonded together at wafer level and cut out by dicing.

In the laminated structure of the upper and lower two chips, the first chip **910** may be an analog chip (sensor chip) including at least one analog component of each pixel circuit, e.g., the photoelectric conversion elements arranged in array form.

For example, the first chip **910** may include only the photoelectric conversion elements of the pixel circuits as described above with reference to the preceding FIGS. Alternatively, the first chip **910** may include further elements of each pixel circuit. For example, the first chip **910** may include, in addition to the photoelectric conversion elements, at least the transfer transistor, the reset transistor, the amplification transistor, and/or the selection transistor of the pixel circuits. Alternatively, the first chip **910** may include each element of the pixel circuit. Alternatively, the first chip **910** may include each element of the pixel circuit and the capacitor control circuit **230** of FIG. **4**.

The second chip **920** may be mainly a logic chip (digital chip) that includes the elements complementing the elements on the first chip **910** to complete pixel circuits and current control circuits. The second chip **920** may also include analog circuits, for example circuits that quantize analog signals transferred from the first chip **910** through the TCVs.

The second chip **920** may have one or more bonding pads BPD and the first chip **910** may have openings OPN for use in wire-bonding to the second chip **920**.

The solid-state imaging device **23020** with the laminated structure of the two chips **910**, **920** may have the following characteristic configuration:

The electrical connection between the first chip **910** and the second chip **920** is performed through, for example, the TCVs. The TCVs may be arranged at chip ends or between a pad region and a circuit region. The TCVs for transmitting control signals and supplying power may be mainly concentrated at, for example, the four corners of the solid-state imaging device **23020**, by which a signal wiring area of the first chip **910** can be reduced.

FIG. **16** shows a possible allocation of elements of a solid-stage imaging device across the first chip **910** and the second chip **920** of FIG. **15**.

The first chip **910** may include the pixel circuits **100** with photoelectric conversion elements. The second chip **920** may include inter alia the column signal processing unit **14** with the constant current circuit **200**. One through contact via **915** per pixel circuit **100** may be part of the data signal line VSL and passes the pixel output signal Vout from the first chip **910** to the second chip **920**.

In addition, a capacitance control circuit **230** connected between the data signal line VSL and the current control capacitor **210** to reduce a capacitive load effective for the pixel output signal Vout may be formed on the first chip **910**.

The pre-charge circuit **240** may include an n channel FET as first switching element **241** that passes the voltage reference potential GND or another low voltage to the data signal line VSL in the pre-charge periods.

The substrate bulk of the transistors of the pixel circuits **100** and the capacitance control circuit **230** may be connected. A further through contact via **915** may pass a bulk potential BLK from the second chip **920** to the first chip **910**. The n channel FET of the pre-charge circuit **240** may be realized on the second chip **920** as illustrated, or on the first chip **910**. In both cases, the substrate bulk of the n channel FET of the pre-charge circuit **240** may be connected to the bulk potential BLK, The source of the n channel FET of the pre-charge circuit **240** may be connected to the voltage reference potential GND as illustrated or to the bulk potential BLK.

The bulk potential BLK may be more negative than the voltage reference potential GND to further expand the voltage swing of the pixel output signal Vout.

FIG. **17** is a block diagram depicting an example of schematic configuration of a vehicle control system as an example of a mobile body control system to which the technology according to an embodiment of the present disclosure can be applied.

The vehicle control system **12000** includes a plurality of electronic control units connected to each other via a communication network **12001**. In the example depicted in FIG. **17**, the vehicle control system **12000** includes a driving system control unit **12010**, a body system control unit **12020**, an outside-vehicle information detecting unit **12030**, an in-vehicle information detecting unit **12040**, and an integrated control unit **12050**. In addition, a microcomputer **12051**, a sound/image output section **12052**, and a vehicle-mounted network interface **12053** are illustrated as a functional configuration of the integrated control unit **12050**.

The driving system control unit **12010** controls the operation of devices related to the driving system of the vehicle in accordance with various kinds of programs. For example, the driving system control unit **12010** functions as a control device for a driving force generating device for generating the driving force of the vehicle, such as an internal combustion engine, a driving motor, or the like, a driving force transmitting mechanism for transmitting the driving force to wheels, a steering mechanism for adjusting the steering angle of the vehicle, a braking device for generating the braking force of the vehicle, and the like.

The body system control unit **12020** controls the operation of various kinds of devices provided to a vehicle body in accordance with various kinds of programs. For example, the body system control unit **12020** functions as a control device for a keyless entry system, a smart key system, a power window device, or various kinds of lamps such as a headlamp, a backup lamp, a brake lamp, a turn signal, a fog lamp, or the like. In this case, radio waves transmitted from a mobile device as an alternative to a key or signals of various kinds of switches can be input to the body system control unit **12020**. The body system control unit **12020** receives these input radio waves or signals, and controls a door lock device, the power window device, the lamps, or the like of the vehicle.

The outside-vehicle information detecting unit **12030** detects information about the outside of the vehicle including the vehicle control system **12000**. For example, the outside-vehicle information detecting unit **12030** is connected with an imaging section **12031**. The outside-vehicle information detecting unit **12030** makes the imaging section **12031** imaging an image of the outside of the vehicle, and receives the imaged image. On the basis of the received image, the outside-vehicle information detecting unit **12030** may perform processing of detecting an object such as a human, a vehicle, an obstacle, a sign, a character on a road surface, or the like, or processing of detecting a distance thereto.

The imaging section **12031** may be or may include an image sensor assembly or a solid-state imaging device with a constant current circuit including a current control capacitor and a ramp generator according to the embodiments of the present disclosure. The light received by the imaging section **12031** may be visible light, or may be invisible light such as infrared rays or the like.

The in-vehicle information detecting unit **12040** detects information about the inside of the vehicle and may be or may include an image sensor assembly or a solid-state imaging device with a constant current circuit including a current control capacitor and a ramp generator according to the embodiments of the present disclosure. The in-vehicle information detecting unit **12040** is, for example, connected with a driver state detecting section **12041** that detects the state of a driver. The driver state detecting section **12041**, for example, includes a camera that includes the solid-stage imaging device and that is focused on the driver. On the basis of detection information input from the driver state detecting section **12041**, the in-vehicle information detecting unit **12040** may calculate a degree of fatigue of the driver or a degree of concentration of the driver, or may determine whether the driver is dozing.

The microcomputer **12051** can calculate a control target value for the driving force generating device, the steering mechanism, or the braking device on the basis of the information about the inside or outside of the vehicle which information is obtained by the outside-vehicle information detecting unit **12030** or the in-vehicle information detecting unit **12040**, and output a control command to the driving system control unit **12010**. For example, the microcomputer **12051** can perform cooperative control intended to implement functions of an advanced driver assistance system

(ADAS) which functions include collision avoidance or shock mitigation for the vehicle, following driving based on a following distance, vehicle speed maintaining driving, a warning of collision of the vehicle, a warning of deviation of the vehicle from a lane, or the like.

In addition, the microcomputer **12051** can perform cooperative control intended for automatic driving, which makes the vehicle to travel autonomously without depending on the operation of the driver, or the like, by controlling the driving force generating device, the steering mechanism, the braking device, or the like on the basis of the information about the outside or inside of the vehicle which information is obtained by the outside-vehicle information detecting unit **12030** or the in-vehicle information detecting unit **12040**.

In addition, the microcomputer **12051** can output a control command to the body system control unit **12020** on the basis of the information about the outside of the vehicle which information is obtained by the outside-vehicle information detecting unit **12030**. For example, the microcomputer **12051** can perform cooperative control intended to prevent a glare by controlling the headlamp so as to change from a high beam to a low beam, for example, in accordance with the position of a preceding vehicle or an oncoming vehicle detected by the outside-vehicle information detecting unit **12030**.

The sound/image output section **12052** transmits an output signal of at least one of a sound or an image to an output device capable of visually or audible notifying information to an occupant of the vehicle or the outside of the vehicle. In the example of FIG. **17**, an audio speaker **12061**, a display section **12062**, and an instrument panel **12063** are illustrated as the output device. The display section **12062** may, for example, include at least one of an on-board display or a head-up display.

FIG. **18** is a diagram depicting an example of the installation position of the imaging section **12031**, wherein the imaging section **12031** may include imaging sections **12101**, **12102**, **12103**, **12104**, and **12105**.

The imaging sections **12101**, **12102**, **12103**, **12104**, and **12105** are, for example, disposed at positions on a front nose, side-view mirrors, a rear bumper, and a back door of the vehicle **12100** as well as a position on an upper portion of a windshield within the interior of the vehicle. The imaging section **12101** provided to the front nose and the imaging section **12105** provided to the upper portion of the windshield within the interior of the vehicle obtain mainly an image of the front of the vehicle **12100**. The imaging sections **12102** and **12103** provided to the side view mirrors obtain mainly an image of the sides of the vehicle **12100**. The imaging section **12104** provided to the rear bumper or the back door obtains mainly an image of the rear of the vehicle **12100**. The imaging section **12105** provided to the upper portion of the windshield within the interior of the vehicle is used mainly to detect a preceding vehicle, a pedestrian, an obstacle, a signal, a traffic sign, a lane, or the like.

Incidentally, FIG. **18** depicts an example of photographing ranges of the imaging sections **12101** to **12104**. An imaging range **12111** represents the imaging range of the imaging section **12101** provided to the front nose. Imaging ranges **12112** and **12113** respectively represent the imaging ranges of the imaging sections **12102** and **12103** provided to the side view mirrors. An imaging range **12114** represents the imaging range of the imaging section **12104** provided to the rear bumper or the back door. A bird's-eye image of the vehicle **12100** as viewed from above is obtained by superimposing image data imaged by the imaging sections **12101** to **12104**, for example.

At least one of the imaging sections **12101** to **12104** may have a function of obtaining distance information. For example, at least one of the imaging sections **12101** to **12104** may be a stereo camera constituted of a plurality of imaging elements, imaging element having pixels for phase difference detection or may include a ToF module including an image sensor assembly or a solid-state imaging device with a constant current circuit including a current control capacitor and a ramp generator according to the embodiments of the present disclosure.

For example, the microcomputer **12051** can determine a distance to each three-dimensional object within the imaging ranges **12111** to **12114** and a temporal change in the distance (relative speed with respect to the vehicle **12100** on the basis of the distance information obtained from the imaging sections **12101** to **12104**, and thereby extract, as a preceding vehicle, a nearest three-dimensional object in particular that is present on a traveling path of the vehicle **12100** and which travels in substantially the same direction as the vehicle **12100** at a predetermined speed (for example, equal to or more than 0 km/hour). Further, the microcomputer **12051** can set a following distance to be maintained in front of a preceding vehicle in advance, and perform automatic brake control (including following stop control), automatic acceleration control (including following start control), or the like. It is thus possible to perform cooperative control intended for automatic driving that makes the vehicle travel autonomously without depending on the operation of the driver or the like.

For example, the microcomputer **12051** can classify three-dimensional object data on three-dimensional objects into three-dimensional object data of a two-wheeled vehicle, a standard-sized vehicle, a large-sized vehicle, a pedestrian, a utility pole, and other three-dimensional objects on the basis of the distance information obtained from the imaging sections **12101** to **12104**, extract the classified three-dimensional object data, and use the extracted three-dimensional object data for automatic avoidance of an obstacle. For example, the microcomputer **12051** identifies obstacles around the vehicle **12100** as obstacles that the driver of the vehicle **12100** can recognize visually and obstacles that are difficult for the driver of the vehicle **12100** to recognize visually. Then, the microcomputer **12051** determines a collision risk indicating a risk of collision with each obstacle. In a situation in which the collision risk is equal to or higher than a set value and there is thus a possibility of collision, the microcomputer **12051** outputs a warning to the driver via the audio speaker **12061** or the display section **12062**, and performs forced deceleration or avoidance steering via the driving system control unit **12010**. The microcomputer **12051** can thereby assist in driving to avoid collision.

At least one of the imaging sections **12101** to **12104** may be an infrared camera that detects infrared rays. The microcomputer **12051** can, for example, recognize a pedestrian by determining whether or not there is a pedestrian in imaged images of the imaging sections **12101** to **12104**. Such recognition of a pedestrian is, for example, performed by a procedure of extracting characteristic points in the imaged images of the imaging sections **12101** to **12104** as infrared cameras and a procedure of determining whether or not it is the pedestrian by performing pattern matching processing on a series of characteristic points representing the contour of the object. When the microcomputer **12051** determines that there is a pedestrian in the imaged images of the imaging sections **12101** to **12104**, and thus recognizes the pedestrian, the sound/image output section **12052** controls the display section **12062** so that a square contour line for emphasis is displayed so as to be superimposed on the recognized pedestrian. The sound/image output section **12052** may also control the display section **12062** so that an icon or the like representing the pedestrian is displayed at a desired position.

The example of the vehicle control system to which the technology according to an embodiment of the present disclosure is applicable has been described above. By applying the an image sensor assembly or a solid-state imaging device with a constant current circuit including a current control capacitor and a ramp generator according to the embodiments of the present disclosure, the sensors can be provided with higher dynamic range and better signal-to-noise ratio.

Additionally, embodiments of the present technology are not limited to the above-described embodiments, but various changes can be made within the scope of the present technology without departing from the gist of the present technology.

The solid-state imaging device according to the present disclosure may be any device used for analyzing and/or processing radiation such as visible light, infrared light, ultraviolet light, and X-rays. For example, the solid-state imaging device may be any electronic device in the field of traffic, the field of home appliances, the field of medical and healthcare, the field of security, the field of beauty, the field of sports, the field of agriculture, the field of image reproduction or the like.

Specifically, in the field of image reproduction, the solid-state imaging device may be a device for capturing an image to be provided for appreciation, such as a digital camera, a smart phone, or a mobile phone device having a camera function. In the field of traffic, for example, the solid-state imaging device may be integrated in an in-vehicle sensor that captures the front, rear, peripheries, an interior of the vehicle, etc. for safe driving such as automatic stop, recognition of a state of a driver, or the like, in a monitoring camera that monitors traveling vehicles and roads, or in a distance measuring sensor that measures a distance between vehicles or the like.

In the field of home appliances, the solid-state imaging device may be integrated in any type of sensor that can be used in devices provided for home appliances such as TV receivers, refrigerators, and air conditioners to capture gestures of users and perform device operations according to the gestures. Accordingly the solid-state imaging device may be integrated in home appliances such as TV receivers, refrigerators, and air conditioners and/or in devices controlling the home appliances. Furthermore, in the field of medical and healthcare, the solid-state imaging device may be integrated in any type of sensor, e.g. a solid-state image device, provided for use in medical and healthcare, such as an endoscope or a device that performs angiography by receiving infrared light.

In the field of security, the solid-state imaging device can be integrated in a device provided for use in security, such as a monitoring camera for crime prevention or a camera for person authentication use. Furthermore, in the field of beauty, the solid-state imaging device can be used in a device provided for use in beauty, such as a skin measuring instrument that captures skin or a microscope that captures a probe. In the field of sports, the solid-state imaging device can be integrated in a device provided for use in sports, such as an action camera or a wearable camera for sport use or the like. Furthermore, in the field of agriculture, the solid-state imaging device can be used in a device provided for use in agriculture, such as a camera for monitoring the condition of fields and crops.

The present technology can also be configured as described below:

(1) An image sensor array, including:

a pixel circuit configured to generate a pixel output signal with an amplitude related to an intensity of detected light and to pass the pixel output signal to a data signal line for a selection period;

a current control capacitor configured to supply a current to the data signal line through a first electrode in the selection period; and

a ramp generator configured to generate a voltage ramp signal and to pass the voltage ramp signal to a second electrode of the current control capacitor in the selection period.

(2) The image sensor array according to (1), further including:

a capacitance control circuit connected between the data signal line and the current control capacitor, wherein the capacitance control circuit is configured to reduce a capacitive load effective for the pixel output signal.

(3) The image sensor array according to (2),

wherein the capacitance control circuit includes an FET.

(4) The image sensor array according to any of (1) to (3), further including:

a pre-charge circuit configured to pre-charge the data signal line in a pre-charge period outside the selection period.

(5) The image sensor array according to (4),

wherein the pre-charge circuit includes a first switching element configured to pass a voltage reference potential to the data signal line in the pre-charge period.

(6) The image sensor array according to any of (4) to (5),

wherein the pre-charge circuit is configured to start each pre-charge period with end of a preceding selection period and to end each pre-charge period with start of a following selection period.

(7) The image sensor array according to any of (1) to (6), further including:

a capacitor control circuit configured to decouple the second electrode of the current control capacitor from an output of the ramp generator in an initial phase at a start of the selection period and/or in a tail phase at an end of the selection period.

(8) The image sensor array according to (7),

wherein the capacitor control circuit includes a second switching element configured to disconnect the second electrode of the current control capacitor from the output of the ramp generator in the initial phase and/or in the tail phase.

(9) The image sensor array according to any of (7) to (8),

wherein the capacitor control circuit includes a third switching element configured to pass a first constant voltage to the second electrode of the current control capacitor in the initial phase.

(10) The image sensor array according to any of (7) to (9),

wherein the capacitor control circuit includes a fourth switching element configured to pass a second constant voltage to the second electrode of the current control capacitor in the trail phase.

(11) The image sensor array according to any of (1) to (10),

wherein the ramp generator includes a plurality of switchable current supply cells connected in parallel and an output resistor, wherein each current supply cell includes a cell current source and a primary switching element connected in series, and wherein the output resistor is connected between the parallel connected current supply cells and a constant voltage.

(12) The image sensor array according to (11),

wherein each switchable current cell further includes a secondary switching element connected in series between the cell current source and the constant voltage.

(13) The image sensor array according to any of (11) to (12), further including:

a counter configured to control the switchable current supply cells.

(14) A solid-state imaging device, including:

a plurality of pixel circuits, wherein each pixel circuit is configured to generate a pixel output signal with an amplitude related to an intensity of detected light, and wherein each pixel circuit is assigned to one of a plurality of pixel columns,

a plurality of data signal lines, wherein each data signal line is configured to receive the pixel output signals of a plurality of pixel circuits assigned to a same pixel column, and wherein to each data signal line one selected pixel circuit passes the pixel output signal for a selection period;

a plurality of current control capacitors, wherein each current control capacitor is configured to supply a current to one of the data signal lines through a first electrode in the selection periods; and

a ramp generator configured to generate a voltage ramp signal and to pass the voltage ramp signal to second electrodes of the current control capacitors in the selection period.

(15) The solid-state imaging device according to (14), further including:

a capacitor control circuit configured to decouple the second electrodes of the current control capacitors from an output of the ramp generator in initial phases at start of the selection periods and/or in tail phases at end of the selection periods.

The invention claimed is:

1. An image sensor array, comprising:

a pixel circuit configured to generate a pixel output signal with an amplitude related to an intensity of detected light and to pass the pixel output signal to a data signal line for a selection period;

a current control capacitor configured to supply a current to the data signal line through a first electrode in the selection period; and

a ramp generator configured to generate a voltage ramp signal and to pass the voltage ramp signal to a second electrode of the current control capacitor in the selection period.

2. The image sensor array according to claim **1**, further comprising:

a capacitance control circuit connected between the data signal line and the current control capacitor, wherein the capacitance control circuit is configured to reduce a capacitive load effective for the pixel output signal.

3. The image sensor array according to claim **2**,

wherein the capacitance control circuit comprises an FET.

4. The image sensor array according to claim **1**, further comprising:

a pre-charge circuit configured to pre-charge the data signal line in a pre-charge period outside the selection period.

5. The image sensor array according to claim **4**,

wherein the pre-charge circuit comprises a first switching element configured to pass a voltage reference potential to the data signal line in the pre-charge period.

6. The image sensor array according to claim **4**,

wherein the pre-charge circuit is configured to start each pre-charge period with end of a preceding selection period and to end each pre-charge period with start of a following selection period.

7. The image sensor array according to claim **1**, further comprising:

a capacitor control circuit configured to decouple the second electrode of the current control capacitor from an output of the ramp generator in an initial phase at a start of the selection period and/or in a tail phase at an end of the selection period.

8. The image sensor array according to claim **7**,

wherein the capacitor control circuit comprises a second switching element configured to disconnect the second electrode of the current control capacitor from the output of the ramp generator in the initial phase and/or in the tail phase.

9. The image sensor array according to claim **7**,

wherein the capacitor control circuit comprises a third switching element configured to pass a first constant voltage to the second electrode of the current control capacitor in the initial phase.

10. The image sensor array according to claim **7**,

wherein the capacitor control circuit comprises a fourth switching element configured to pass a second constant voltage to the second electrode of the current control capacitor in the trail phase.

11. The image sensor array according to claim **1**,

wherein the ramp generator comprises a plurality of switchable current supply cells connected in parallel and an output resistor, wherein each current supply cell comprises a cell current source and a primary switching element connected in series, and wherein the output resistor is connected between the parallel connected current supply cells and a constant voltage.

12. The image sensor array according to claim **11**,

wherein each switchable current cell further comprises a secondary switching element connected in series between the cell current source and the constant voltage.

13. The image sensor array according to claim **11**, further comprising:

a counter configured to control the switchable current supply cells.

14. A solid-state imaging device, comprising:

a plurality of pixel circuits, wherein each pixel circuit is configured to generate a pixel output signal with an amplitude related to an intensity of detected light, and wherein each pixel circuit is assigned to one of a plurality of pixel columns,

a plurality of data signal lines, wherein each data signal line is configured to receive the pixel output signals of a plurality of pixel circuits assigned to a same pixel column, and wherein to each data signal line one selected pixel circuit passes the pixel output signal for a selection period;

a plurality of current control capacitors, wherein each current control capacitor is configured to supply a current to one of the data signal lines through a first electrode in the selection periods; and a ramp generator configured to generate a voltage ramp signal and to pass the voltage ramp signal to second electrodes of the current control capacitors in the selection period.

15. The solid-state imaging device according to claim **14**, further comprising:

a capacitor control circuit configured to decouple the second electrodes of the current control capacitors from an output of the ramp generator in initial phases at start of the selection periods and/or in tail phases at end of the selection periods.

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